Supplement to "Radiative cancellations and localized linear modes in three-dimensional sextic Q-balls"

Finn Hinrichsen

29 October 2026 — Preprint 0.52

This Supplement records the nonlinear coefficient's dependency audit, followed by exploratory driving, formation and numerical-control material retained from the earlier manuscript. The complete linear proof lemmas are in Appendix A of the main paper. The latter material is not evidence for additional localized linear modes. Equation and section references without an S prefix refer to the main paper. The companion archive is distributed with release `qball-bic-v0.51` (see the data availability statement of the main paper); text, data and certificates are licensed under CC BY 4.0, code under the MIT License. Historical records are preserved. The 0.51 preprint repeated the three primary linear certificate runs with unchanged proof code; it did not repeat the exploratory physical evolutions below. Preprint 0.52 adds explanatory material only.

Contents

S1 Dependency audit for the monopole coefficient **1**
S1.1 Status of the five dependencies . . . 2
S1.2 Exterior reduction and unique outgoing response . . . 3
S1.3 Global adjoint weights and the error identity . . . 3
S1.4 Weighted regularity and the continuation assumption . . . 4

S2 Stored numerical response and figure diagnostics **4**
S2.1 Numerical second-harmonic response of the dipole mode . . . 4
S2.2 Radial nonlinear diagnostics . . . 5
S2.3 Spatial mode shapes and phase-resolved reconstruction . . . 6

S3 Dimensional and discretization comparisons **8**
S3.1 Exploratory sensitivity to the numerical lattice . . . 8

S4 Prescribed driving and response diagnostics **10**

S5 Exploratory concentration and constrained transport **27**

S6 Separate numerical-method controls **38**

S7 Archive structure and evidential scope **38**

S1 Dependency audit for the monopole coefficient

The five hypotheses in the main text describe dependencies of the coefficient enclosure. The scalar summation alone does not prove its input lemmas. This register distinguishes inherited computer-assisted results, analytic implications, conventions and the additional regularity assumption on a hypothetical nonlinear family. The accompanying archive is distributed with release `qball-bic-v0.51` (see the data availability statement of the main paper).

S1.1 Status of the five dependencies

1. **Common exact background and mode: inherited existence result.** The defining C2 certificate, `ZERT-T2-A-L36.json` in `archive/proofs/C1-C2/`, encloses one common exact tuple $z_* = (a_*, c_*, \rho_*, \omega_*)$. The analytic implication is the linear theorem together with the origin, tail and inclusion lemmas in Section A. All following enclosures refer to this same tuple. Neither a distinct fit nor every point in the parameter box is assumed to be an exact root. The code and arithmetic trust assumption is inherited; uniqueness of the tuple is unnecessary.

2. **Regular/Jost identification: inherited Wronski argument.** At this tuple, the two exact matching Wronskians vanish, so the normalized decaying Jost vector belongs to the two-dimensional regular origin span. Section A.5 supplies this analytic step. The archived matching-component bounds enclose its coefficients in that span. Their residual intervals containing zero would not, by themselves, establish a common solution. The exterior evaluation binds the C2 box and verifies the rational majorants

$$|f_*(r)| \le \frac{97}{8}\frac{e^{-0.495r}}{r}, \qquad |u(r)| + |v(r)| \le 1.097e^{-0.287r}, \qquad r \ge 36. \tag{S1}$$

   These are the fundamental mode's reduced components and Jost normalization, not unit-L^2 data.

3. **Source and normalization: definition plus checked binding.** The exact source is the full quadratic product in the main text. The fixed nominal unit-L^2 mode is divided by $s_0 = 2558188928468177/2251799813685248$, and quadratic quantities by s_0^2. This exact rational conversion defines the comparison data; it does not set the exact mode's L^2 norm equal to s_0. The source and response exports use the same nominal functions and carrier conjugation. Exact-to-nominal errors remain in the budget. This item is a convention and input-identity check, rather than a new existence assumption. The archive's normalization and error-identity lemmas specify it independently of the stored scalar sum.

4. **Exact outgoing problem: analytic implication with checked inputs.** The reduced second-harmonic operator has a real symmetric locally bounded potential and both $q_\pm$ strictly between 5/2 and 23/5. These inequalities are checked over the same exact C2 box. The exterior evaluation gives, at $R = 64$,

$$\int_{64}^{\infty} \|V_*\|_2 \, \mathrm{d}r < 10^{-28}, \qquad \int_{64}^{\infty} \|F_*\|_2 \, \mathrm{d}r < 10^{-32}. \tag{S2}$$

   In particular the contraction condition $\int_R^\infty \|V_*\|_2/q_{\min} < 1/2$ is verified at the exact frequencies. The construction below includes the exterior source and proves the unique regular outgoing problem and its adjoint extraction. Thus this item is not an untested numerical Robin-condition assumption. It still depends on the inherited exact-mode and tail lemmas and the correctness of their validated evaluation.

5. **Quantitative bounds and complete budget: checked inputs and sum.** The complete-potential enclosure gives $\int_0^\infty \|V_*\|_2 \, \mathrm{d}r < 25/4$. The analytic argument below therefore supplies $\|Z\|_2 < 26/5$, $\|Z'\|_2 < 14$ for the exact adjoint. The archived component budget bounds source and operator errors, the actual reconstructed response residual, origin terms, matching jumps, exterior corrections and coefficient extraction. Its classification is a satisfied conditional coefficient criterion; there is no unfilled numerical summand in this $L = 0$ budget. Accepting the historical input chain and its directed-arithmetic execution remains a trust assumption, rather than a new independent verification performed for this manuscript revision.

All five items have identified analytical or computational support. The adjective *conditional* records this inherited proof chain and its software assumptions; it does not designate an omitted term of the reported scalar budget. The separate assumption that a hypothetical nonlinear family is twice continuously differentiable in a fixed weighted space restricts the family excluded by the continuation argument. The circular-polarization case and a quantitative quadrupole nonzero certificate remain outside this monopole conclusion.

S1.2 Exterior reduction and unique outgoing response

Let $-U'' - Q^2U + VU = F$, with $Q = \operatorname{diag}(q_+, q_-)$ and $q_j \geq q_{\min} > 0$. In the computational convention both outgoing carriers are $e^{iq_j r}$. Define

$$(Kg)(r) = \int_r^\infty Q^{-1} \sin(Q(s-r)) g(s)\, \mathrm{d}s, \quad E(r) = e^{iQr}, \quad H = E + KVH, \quad p = KVp - KF. \tag{S3}$$

Set $A_R = \int_R^\infty \|V\|_2$, $B_R = \int_R^\infty \|F\|_2$, $\eta = A_R/q_{\min} < 1/2$, $\delta = \eta/(1-\eta)$ and $\gamma_R = 1/(1-\delta)$. Neumann inversion on bounded functions gives

$$\|H\| \leq \frac{1}{1-\eta}, \quad \|H - E\| \leq \delta, \quad \|H' - E'\| \leq \frac{A_R}{1-\eta}, \quad \|p\| \leq \frac{B_R}{q_{\min}(1-\eta)}, \quad \|p'\| \leq \frac{B_R}{1-\eta}. \tag{S4}$$

The corrections and their first derivatives vanish at infinity. Since $E(R)$ is unitary and $\delta < 1$, $H(R)$ is invertible with inverse norm at most γ_R. Thus $U = HC + p$ has the exact maps

$$\begin{aligned} T = H(R)^{-1}, \quad c = -Tp(R), \quad D = H'(R)T, \quad b = p'(R) - Dp(R), \\ C = TU(R) + c, \qquad U'(R) = DU(R) + b. \end{aligned} \tag{S5}$$

The source contributes both b and c. Dropping it would change the problem even if the homogeneous maps were exact. For $d_R = [A_R/(1-\eta) + q_{\max}\delta]\gamma_R$, the bounds are

$$\|T - E(R)^{-1}\| \leq \delta\gamma_R, \quad \|D - iQ\| \leq d_R, \quad \|c\| \leq \gamma_R\|p\|, \quad \|b\| \leq \|p'\| + (q_{\max} + d_R)\|p\|. \tag{S6}$$

They follow by inserting $H - E$ into the inverse identity and into $H' - iQH$; all matrix products keep their displayed order.

For real symmetric V, the conserved matrix current is $(H^\dagger H' - H'^\dagger H)/(2i) = Q$. Consequently

$$\frac{D - D^\dagger}{2i} = H(R)^{-\dagger} Q H(R)^{-1} > 0. \tag{S7}$$

A homogeneous regular solution $U(0) = 0$ satisfying $U'(R) = DU(R)$ has real interior Green energy. Its boundary imaginary part is positive unless $U(R) = 0$. Hence both Cauchy data vanish and $U = 0$. The finite regular-to-boundary matrix is therefore invertible, which proves existence and uniqueness for the inhomogeneous boundary data. The adjoint problem uses $D^\dagger$ and has negative definite boundary imaginary part, so the same argument proves its uniqueness. This reasoning applies to the two open second-harmonic channels. The fundamental embedded mode has a closed channel and does not satisfy this positivity premise.

S1.3 Global adjoint weights and the error identity

Let $A(r) = \int_r^\infty \|V(s)\|_2\, \mathrm{d}s$ and $h(r) = e^{A(r)/q_{\min}}$. Ordered Volterra iteration on the whole half-line gives outgoing and incoming matrices $H_\pm$ with

$$\|H_\pm(r)\|_2 \leq h(r), \qquad \|H'_\pm(r)\|_2 \leq q_{\max} + q_{\min}(h(r) - 1). \tag{S8}$$

No smallness condition on $A(0)$ is required: the nth integral has bound $(A(r)/q_{\min})^n/n!$. The conserved sesquilinear Wronskians are $W(H_\pm, H_\pm) = \pm 2iQ$, with zero cross-pairings. If $\Phi(0) = 0$,

$\Phi'(0) = I$, then $Z = \Phi H_+(0)^{-\dagger}$ obeys $W(H_+, Z) = I$ and $W(Z, Z) = 0$. Invertibility of $H_+(0)$ follows directly from its positive current. Writing $Z = H_+C_+ + H_-C_-$ gives $C_+ = (2iQ)^{-1}$ and $C_-^\dagger QC_- = (1/4)Q^{-1}$. It follows that $\|C_\pm\|_2 \le 1/(2q_{\min})$ and therefore

$$\|Z(r)\|_2 \le \frac{h(r)}{q_{\min}}, \qquad \|Z'(r)\|_2 \le \frac{q_{\max}}{q_{\min}} + h(r) - 1. \tag{S9}$$

The checked $A(0) < 25/4$, $q_{\min} = 5/2$, $q_{\max} = 23/5$ give $h(r) < e^{5/2} < 13$ and the weights 26/5, 14. The rational estimate $e \le 87/32 < 11/4$, $\sqrt{e} < 5/3$ yields $e^{5/2} < 605/48 < 13$; it uses the exponential series remainder after $n = 3$, bounded by $(1/24)/(1 - 1/5)$.

For the fixed piecewise smooth nominal response $\widehat{U}$, define $I_F = \int_0^R \|F_* - F_n\|_2$, $I_W = \int_0^R \|W_* - W_n\|_2\|\widehat{U}\|_2$, $I_R = \int_0^R \|L_n\widehat{U} - F_n\|_2$, $J_U = \sum \|[\widehat{U}]\|_2$, $J_D = \sum \|[\widehat{U}']\|_2$, $U_0 = \|\widehat{U}(0)\|_2$ and $U_R = \|\widehat{U}(R)\|_2$. Integrating Green's identity on each smooth piece keeps the origin, jump and exterior boundary terms. Bounding each by the exact adjoint weights yields

$$\|C_* - C_{\mathrm{num}}\|_2 \le \frac{26}{5}(I_F + I_W + I_R + J_D + B) + 14(J_U + U_0) + O, \tag{S10}$$

where $B = \|b_*\|_2 + \epsilon_D U_R + \|b_n\|_2$ and $O = \epsilon_T U_R + \|c_*\|_2 + \|d_{\mathrm{out}}\|_2$. Here ϵ_D, ϵ_T bound exact-to-nominal maps, b_n is the nominal response's actual boundary mismatch and d_{out} is its extraction difference from the fixed comparison coefficient. W denotes the full reduced operator potential, including the frequency part. Thus frequency errors are retained in I_W, and exterior source terms occur in b_*, c_* exactly once. The archived component result encloses every term of (S10); its positive second-channel margin is the one quoted in the main text.

S1.4 Weighted regularity and the continuation assumption

The already derived weighted-regularity lemma permits $\gamma = 1/10$. Here is its logical link to the exact data, without a new numerical Sobolev-norm calculation. From (S1), write $a = 0.495$, $b = 0.287$. The profile equation gives $(r^2 f')' = r^2 N(f)$ with integrable right-hand side; hence

$$f'(r) = \frac{\ell_\infty}{r^2} - r^{-2}\int_r^\infty s^2 N(f(s))\,\mathrm{d}s. \tag{S11}$$

Integration yields $f(r) = -\ell_\infty/r + O(e^{-ar})$; the known exponential bound forces $\ell_\infty = 0$, so $f' = O(e^{-ar})$. For the fundamental mode $y = (u, v)$, the bounded exterior matrix in $y'' = M(r)y$ gives $y'' = O(e^{-br})$. Its integrability gives a limit of y', which must be zero because $y \to 0$. Thus $y' = O(e^{-br})$. Twice differentiating these actual ODEs propagates the same rates through order four. This differentiates equations, not a modulus inequality. The origin expansions in Section A.5 give smooth Cartesian fields; the compact intermediate region is smooth. Cartesian derivatives through order four are finite sums of radial derivatives times bounded angular factors and inverse powers of r. After multiplication by $e^{\gamma\sqrt{1+r^2}}$, they remain square integrable because $a - \gamma > 0$, $b - \gamma > 0$. Thus the background and every finite complex combination of its three dipole tangents belong to $H^4_\gamma(\mathbb{R}^3)$.

What remains an explicit assumption is that a hypothetical family is C^2 in amplitude in $C^2(\mathbb{T}; H^4_\gamma)$ with this one fixed positive weight, smooth real frequencies and the specified first harmonic tangent (possibly plus a static first tangent). The nonzero coefficient excludes that class. Pointwise smoothness at each radius, changing weights along the family, other first harmonics or circular polarization are not covered by this monopole argument.

S2 Stored numerical response and figure diagnostics

S2.1 Numerical second-harmonic response of the dipole mode

We evaluated the C2 forcing in both $L = 0$ and $L = 2$ using separately reconstructed numerical profiles near the certified point. The reduced real mode components obey $\int_0^\infty (u^2 + v^2)\,\mathrm{d}r = 1$,

including an approximate tail contribution, with $\int Y_{10}^2 \, \mathrm{d}\Omega = 1$. The solver uses the complex conjugate of our positive-carrier convention: both computational unknowns have positive-q outgoing Hankel conditions, and the second unknown is conjugated to reconstruct the physical negative sideband. The power is invariant under this convention change.

Three prescribed settings use exterior radii $48, 64, 64$, source spacings $0.02, 0.02, 0.01$, and relative ODE tolerances $2 \times 10^{-10}, 2 \times 10^{-10}, 2 \times 10^{-11}$. Two further solves at the finest settings use the coarser reconstruction level from the same input algorithm, with different exterior radius and tolerance. This is an input-sensitivity comparison, not an independent implementation. All four outgoing coefficients remain nonzero under these changes; the largest observed relative complex-amplitude change is below 7.8×10^{-8}. This sensitivity is not a rigorous error bound.

At the finest setting the harmonically averaged power coefficients are

Angular sector	Positive sideband	Negative sideband
$L = 0$	2.4239×10^{-7}	6.0359×10^{-5}
$L = 2$	4.4956×10^{-7}	9.1587×10^{-5}

The leading power is ϵ^4 times the sum of these coefficients in the stated normalization. Both physical sidebands carry positive outward energy with the stated outgoing conditions. These values neither determine a lifetime nor measure a nonlinear time-evolution loss. They must not be compared directly with coefficients using the earlier radial maximum-amplitude normalization.

This calculation supports second-harmonic leakage for the approximate C2 input, beyond merely finding open channels and nonzero local sources. It does not certify leakage at the exact enclosed BIC. Float rounding places some input parameters outside the tiny certified box; sampled scaled Hermite-interpolant ODE defects have a sampled maximum of about 5.2012×10^{-6} (rounded upward to 5.3×10^{-6}). These are midpoint samples on 0.01-spaced grids extending to $r = 80, 96$, excluding $|r - 8| \le 0.011$ around the splice, not uniform residual bounds. Differences between reconstruction levels are much smaller. Omitted exterior source and potential contributions were diagnosed, not converted into propagated amplitude-error bounds in that calculation. The separate validated monopole transfer in Section 7.5 addresses these errors using a fixed comparison function and the exact mode's Jost normalization; it does not certify the numerical power table.

A subsequent fixed-coefficient reconstruction checked the source–current identity for each angular sector. For the real symmetric forced operator, $J = \mathrm{Im}(U^\dagger U')$ satisfies $J' = \mathrm{Im}(F^\dagger U)$, where U denotes the two reduced response components and F their forcing. Both computational channels use positive-wavenumber outgoing waves. Keeping the saved matching coefficients fixed, an independent volume quadrature agrees with $\sum_j q_j |C_j|^2$ to relative residuals 9.6×10^{-11} for $L = 0$ and 1.3×10^{-11} for $L = 2$. This is a numerical consistency check of the finite forced problem, not an energy-power measurement or a validated error bound. It does not independently validate a shared source formula. The change between the two volume quadratures is 1.9×10^{-9} or smaller relative to the respective outgoing current; this observation is not a certified quadrature-error estimate.

S2.2 Radial nonlinear diagnostics

Separate computations already address the earlier $n = 1$, $l = 0$ mode near $\omega^2 = 0.7976767871$ and $\rho = 1.7446175448$. They concern the same sextic field equation, not C1 or C2. A finite-amplitude radial evolution compared the nonlinear field with its background and linear response on two spatial grids. For amplitudes $0.01, 0.02, 0.04$, the integrated quadratic flux of the nonlinear remainder increased by factors approximately 15.97 and 15.89 upon doubling amplitude. This supports fourth-order scaling over the tested range; that remainder flux is not an exact measurement of the total lost core energy.

A subsequent harmonic diagnostic identified the dominant laboratory sideband $E = \omega - 2\rho \simeq -2.59611$ as outgoing at two detector radii, on both grids and in both prescribed time windows.

The weaker $E = \omega + 2\rho$ component was resolved but did not satisfy every outgoing-wave criterion. Large residuals in the full field's harmonic fit remained, so these observations do not establish a periodic multifrequency solution.

A separately solved frequency-domain problem, using the same background and mode data but no fitted amplitude from the time evolution, predicted the dominant outgoing coefficient. At $\epsilon = 0.02$, its predicted amplitude was 5.1563×10^{-5}, compared with measured values approximately $(5.03\text{–}5.10) \times 10^{-5}$. The prescribed comparison criteria passed. Shared input profiles make this a separate mechanism test, not a wholly independent reconstruction of the background.

An interval calculation further separated one necessary overlap from zero for an explicitly frozen finite operator on $[0, 44]$:

$$I_2 \in [-0.16383470526432667, -0.10547329373338812]. \tag{S12}$$

The corresponding I_1 interval contained zero, and the original test requiring both exclusions remained failed. Nevertheless, one certified nonzero overlap suffices to violate the necessary simultaneous condition $I_1 = I_2 = 0$ for that frozen problem. Its binary64 spline inputs are treated as exact data, with a free exterior and zero source beyond radius 44. This is not a certificate that the exact continuum Q-ball and its BIC have that overlap: a validated link to the continuum inputs and exterior tail remains required. The linear existence certificates do not, by themselves, supply that quantitative link.

If a leading second-order outgoing amplitude is nonzero, its power is of order ϵ^4. A lifetime law additionally requires a controlled relation to mode energy and slow amplitude evolution, and separation of initial adjustment from persistent radiation. Neither the finite-time observations nor the frozen-operator certificate provides a general nonlinear stability or lifetime theorem. Earlier real-field shell-model normal forms describe a different model and supply no substitute for these Q-ball source and overlap conditions.

S2.3 Spatial mode shapes and phase-resolved reconstruction

The three-dimensional shape is obtained from the computed radial functions and the specified spherical harmonic, not from a point-particle picture. For real radial components and the positive-carrier convention of this paper, we display the rotating-frame perturbation per unit small amplitude,

$$\psi_1(\mathbf{x}, t) = Y_{l0}(\widehat{\mathbf{x}}) \left[\frac{A(r)}{r} e^{i\rho t} + \frac{B(r)}{r} e^{-i\rho t} \right]. \tag{S13}$$

The full first-order field is $e^{i\omega t}(f + \epsilon\psi_1)$. Thus a stationary background density does not mean that the complex field is static. A localized nonzero oscillation can have zero linear radiation; this is distinct from removing the oscillation itself.

Figures S1 and S2 use stored numerical reconstructions near C0 and C2. They are not pointwise interval-certified eigenfunctions and are not a nonlinear three-dimensional time evolution. The C2 arrays were produced in the conjugate carrier convention; the time factors in Eq. (S13) conjugate that convention consistently. Interpolation is used only for display. All lengths are dimensionless model coordinates; no particle diameter or laboratory length is assigned.

A spatial node of ψ_1 is not the BIC cancellation condition. The latter is a condition on the frequency and background parameters. To visualize this distinction, Fig. S3 shows an archived C0 search using a normalized backward-Jost origin residual $F = (F_1, F_2)$. Its complex representation $F_1 + iF_2$ is a device for tracking the winding of two real boundary residuals, not the phase of the physical field. The finite-radius numerical diagnostic predates and does not replace the existence enclosure in Section 4.1.

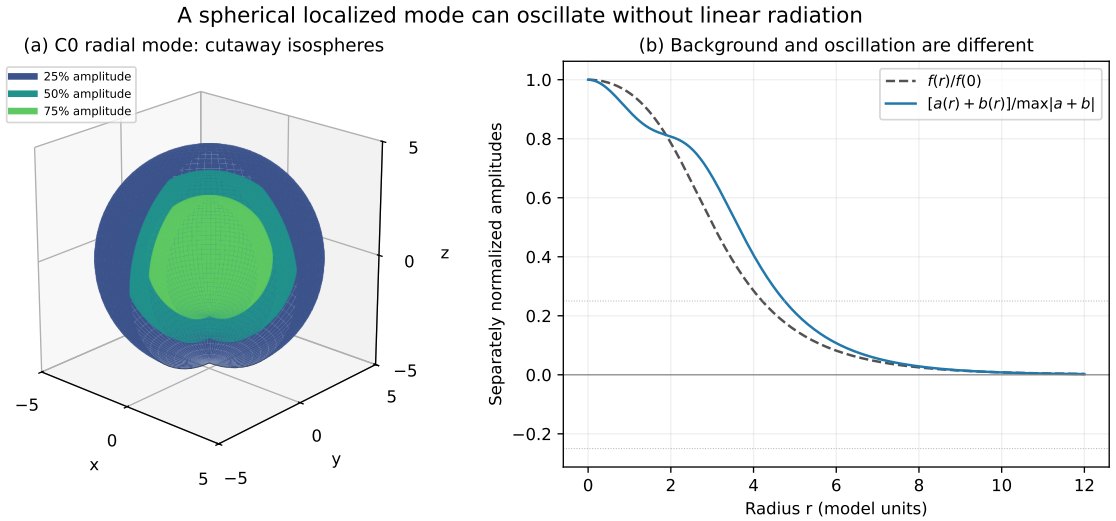


Figure S1: C0 spherical mode at phase zero. Left: cutaway surfaces of constant $|\psi_1|$ at 25%, 50% and 75% of its displayed maximum; color identifies the amplitude level. Removing a quadrant exposes the radial shells and is only a drawing choice. Right: the background and the radial mode, separately normalized for comparison. The shells are level sets, not material surfaces. These are numerical reconstructions associated with C0, not new existence certificates.

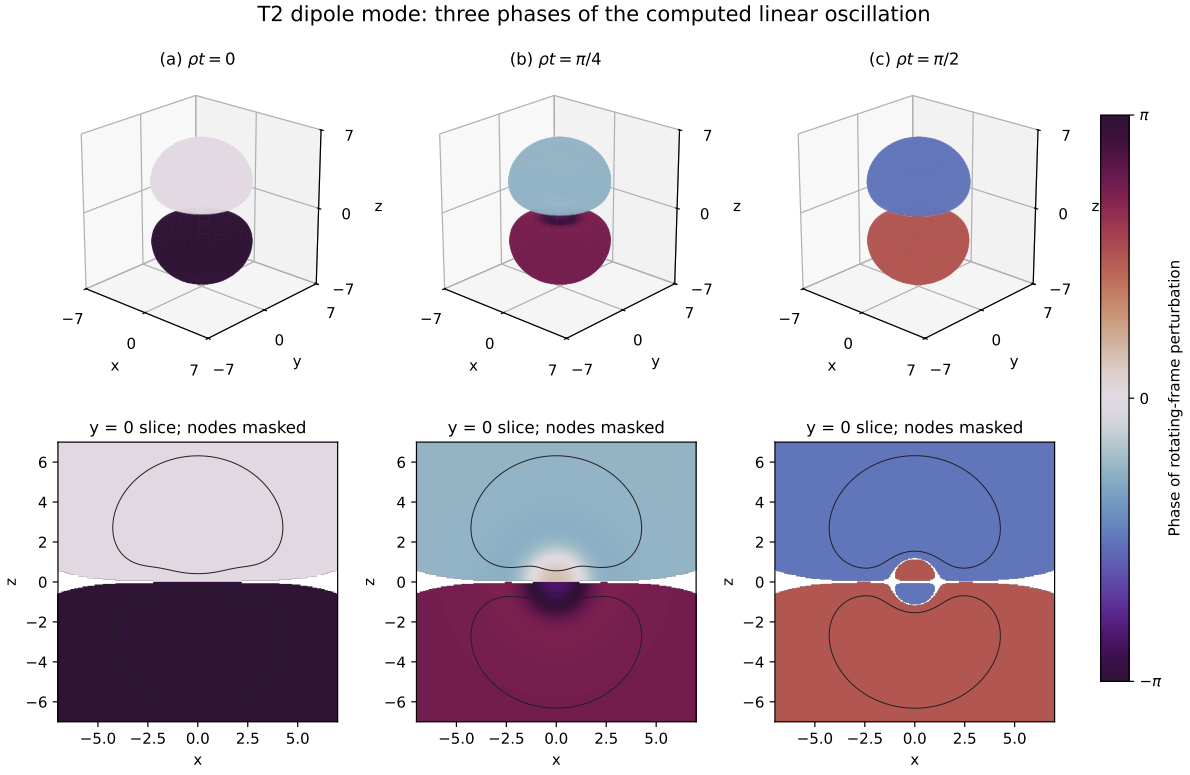


Figure S2: C2 with a real $Y_{10} = \sqrt{3/(4\pi)}z/r$ angular factor. Top: three phases of the linear mode at a common amplitude-isosurface level. Bottom: the corresponding $y = 0$ sections; phase is masked below 1.5% of the common displayed peak, because phase is undefined at a node. Color encodes $\arg \psi_1$ in the rotating frame, with a fixed common gauge and cyclic scale; it is neither a vector direction nor an additional charge. The opposite lobes are in antiphase. The changing quadratures follow the two computed radial sidebands and their analytic time factors, rather than a fresh nonlinear simulation.

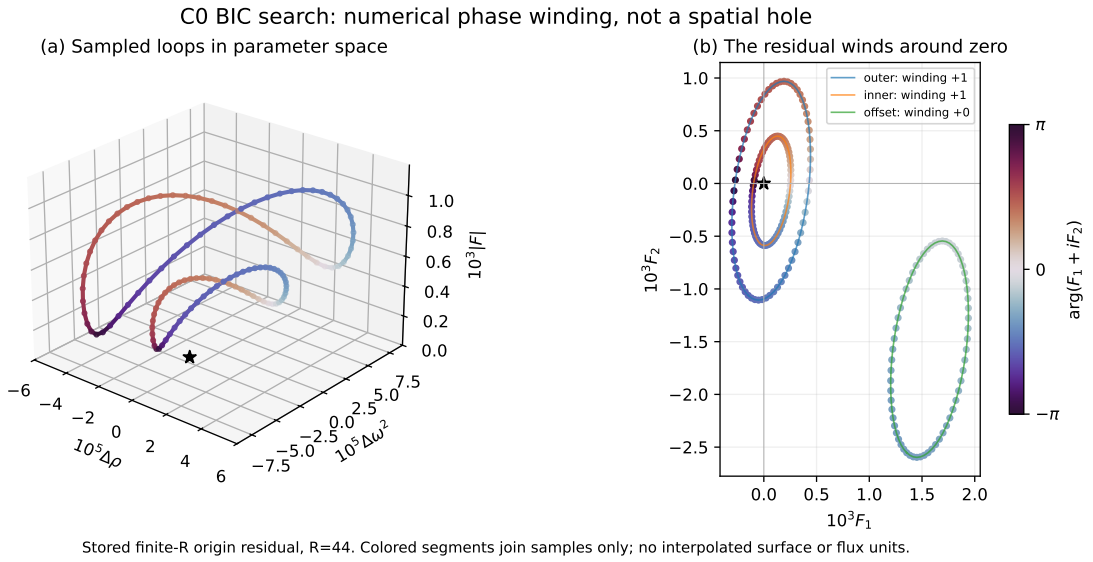


Figure S3: The C0 cancellation in parameter space. Left: two sampled parameter loops lifted by $|F|$, colored by $\arg(F_1 + iF_2)$; the star marks the numerical root. The reference coordinates are $\rho_{\mathrm{ref}} = 1.744625038252254$ and $\omega^2_{\mathrm{ref}} = 0.7976953125$. Right: their residual images and an offset control, with numerical windings $+1, +1, 0$. Lines connect stored samples; no two-dimensional residual surface has been inferred from these loops. The historical calculation uses outer radius 44. The plotted residual is neither radiated power nor an absolute outgoing-wave amplitude, and the third plotting coordinate is not a third physical-space dimension.

S3 Dimensional and discretization comparisons

An exploratory two-dimensional follow-up found further numerical candidates near $\omega^2 = 0.529265$ and 0.525780 after refining the frequency sampling. The original decision rule remained inconclusive; the resolved numerical winding for the former candidate and the sign-change/linewidth evidence for the latter came from adaptively specified follow-up runs. The last estimated inverse-detuning steps are about 4.62, compared with about 2.305 in the three-dimensional sequence. This is not a dimensional scaling theorem. A similar sampling limitation in the earlier three-dimensional search remains a hypothesis.

S3.1 Exploratory sensitivity to the numerical lattice

Exploratory two-dimensional calculations at the seventh radial location also indicate sensitivity to numerical spatial discretization. Here the width is the amplitude decay rate $\Gamma = -\operatorname{Im}\rho$ for the convention $\exp(-i\rho t)$. The reported minimum widths scale approximately as h^4 for a five-point square stencil and h^8 for both a weighted nine-point square stencil and a seven-point triangular stencil. The leading angular radiation components are $\cos(4\theta)$ for both square stencils and $\cos(6\theta)$ for the triangular stencil. These observations are consistent with a perturbative interpretation in which the leading anisotropic operator correction induces a radiation amplitude linear in its strength, and the width is quadratic in that amplitude. The analogous quadratic asymmetry law is established for optical quasi-BICs by Koshelev et al. [19], Eqs. (2)–(3); its application here remains a model-specific interpretation. The reported exponent uncertainties are fit errors, and comparisons of absorbing layers and shifted eigenvalue solves are sensitivity diagnostics rather than certified error bounds. The calculations concern one two-dimensional mode and do not establish three-dimensional or nonlinear persistence. They motivate explicit stencil and resolution checks before angular radiation patterns are interpreted as physical structure.

An additional radiation channel. A separate leading-order probe adds a neutral scalar through $\mathcal{L}_{\rm int} = -g\chi|\phi|^2$. On three fixed radial profiles, the finite-grid form factor $I(k) = \int_0^\infty f(a+b)j_0(kr)r^2\,dr$, $k = \sqrt{\rho^2 - m_\chi^2}$, had respectively one, one and two detected sign-changing zeros in the sampled interval $0 < k < \rho$. These are candidate Born-amplitude cancellations, not a complete count of zeros of the exact integral. The coupling already sources a static neutral field through $(-\Delta + m_\chi^2)\chi_0 = -gf^2$; thus the unchanged Q-ball with $\chi = 0$ is not a solution of the coupled theory. Background and mode backreaction must be included before claiming a coupled bound state. Closing this channel with a strict threshold gap removes propagation at this harmonic, but does not by itself preserve the old channel cancellation or prohibit radiation at higher harmonics.

S4 Prescribed driving and response diagnostics

Linear coupling to the localized mode. For radiation of the same scalar field, a conditional projection identity makes the distinction between scattering and local forcing precise. In a fixed real $l = 1$ harmonic, complexify the doubled reduced perturbation $\psi = (rz_l, r\bar{z}_l)^T$ and write an additive source as S. Equation (21) and its conjugate give

$$\psi_{tt} + 2i\omega\sigma_3\psi_t + \mathcal{K}\psi = S, \qquad \mathcal{K} = -\partial_r^2 + \frac{2}{r^2} + (D - \omega^2)I + C\sigma_1, \tag{S14}$$

where D, C are those of Eqs. (4)–(5), $\sigma_3 = \mathrm{diag}(1, -1)$, and $\sigma_1 = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. The physical perturbation is recovered by the conjugate frequency pairing. For a physical additive source in this real harmonic block, $S_2 = \overline{S_1}$; independent components are used only in the formal complexification, with the conjugate pairing restored for physical forcing. Use the radial inner product with measure dr and the regular self-adjoint domain of $\mathcal{K}$. With $X = (\psi, \psi_t)^T$, define

$$\mathcal{G} = \begin{pmatrix} 0 & I \\ -\mathcal{K} & -2i\omega\sigma_3 \end{pmatrix}, \qquad \mathcal{J} = \begin{pmatrix} 2\omega\sigma_3 & -iI \\ iI & 0 \end{pmatrix}. \tag{S15}$$

Then $\mathcal{G}^\dagger\mathcal{J} + \mathcal{J}\mathcal{G} = 0$ on that domain. For a nonzero localized fundamental mode $H(\rho)u = 0$, where $H(\rho) = \mathcal{K} - \rho^2 I - 2\omega\rho\sigma_3$, put $X_u = (u, i\rho u)^T$. If $\rho > |\omega|$, its conserved form is positive:

$$N = \langle X_u, \mathcal{J}X_u\rangle = -\langle u, H'(\rho)u\rangle = 2\langle u, (\rho I + \omega\sigma_3)u\rangle > 0. \tag{S16}$$

Here ρ is the fundamental frequency, not 2ρ. Positivity on this mode is neither a linear nor a nonlinear stability theorem for the full system. The corresponding left projection $c = \langle X_u, \mathcal{J}X\rangle/N$ obeys

$$\dot{c} = i\rho c - \frac{i}{N}\langle u, S\rangle, \tag{S17}$$

provided the radial Green boundary form vanishes. On $[0, R]$ the additional term is $i[u'^\dagger\psi - u^\dagger\psi']_0^R/N$. At the origin this follows from regular values $u, \psi = O(r^2)$ and regular derivatives $u', \psi' = O(r)$. At infinity the localized mode and its derivative must decay against the controlled scattering field and its derivative; this includes suitable finite wave packets.

Thus a homogeneous linear solution ($S = 0$) with $c(0) = 0$ cannot populate this projection on the fixed background. Spatial separation of an incoming packet alone does not imply exact orthogonality, because the localized mode has tails. A prescribed local additive source can instead have $\langle u, S\rangle \neq 0$ and drive the mode. For $c(0) = 0$ and a finite square-integrable projected source, Eq. (S17) gives

$$N|c(T)|^2 \leq \frac{T}{N}\int_0^T |\langle u, S(t)\rangle|^2\, dt. \tag{S18}$$

This is a mode-projection bound, not a statement about the energy supplied by an unspecified environment. Although Eq. (S16) is positive on this eigenspace, $\mathcal{J}$ is indefinite on the full phase space. No globally passive port model, loading rate, nonlinear formation mechanism, or indefinite maintenance follows. The local source and its physical coupling must be specified separately; the nonlinear radiation obstruction in Section 7 addresses a different question.

A prescribed local pulse. For the exact C2 mode, write $u = (A, B)^T$ with the reduced real components of Eq. (3), in the exact-mode Jost normalization specified in Section 7.5. The projection identity gives a concrete excitation test for this already existing bound mode. In dimensionless mass-one units, prescribe a source with real angular profile Y_{10}. Choose the radial envelope $\chi(r) = (r - 1)^2(2 - r)^2$ on $1 \leq r \leq 2$ and zero elsewhere, and temporal envelope $\eta(t) = \sin^2(\pi t)$ on $0 \leq t \leq 1$ and zero otherwise. The physical source in the original field equation is $J_\phi = \epsilon\chi Y_{10}\eta\exp[i(\omega + \rho)t]$, with real ϵ, the small strength of the external drive in this

linear-response calculation. It is added to the right-hand side of Eq. (20). The reduced source in Eq. (S14) is $S = \epsilon r\chi\eta(e^{i\rho t}, e^{-i\rho t})^T$; its conjugate channel is not independent and must be retained. For the fixed real radial mode components A, B, define

$$I_A = \int_1^2 r\chi(r)A(r)\,dr, \qquad I_B = \int_1^2 r\chi(r)B(r)\,dr.$$

The resulting finite-pulse overlap is

$$L(0) = \frac{I_A}{2} + I_B \int_0^1 \eta(t)e^{-2i\rho t}\,dt.$$

With $N > 0$ from Eq. (S16), vanishing initial projection and vanishing Green boundary form, the pulse leaves $c(1) = -i\epsilon e^{i\rho}L(0)/N$. Exact integration of the stored nominal polynomials, combined with their certified componentwise error bounds and the same certified frequency box, gives an enclosure separated from zero. Conditional on the same exact C2 mode identification and normalization, a conservative bound is $|L(0)| > 1/500$. This certifies nonzero linear mode projection for nonzero ϵ, vanishing initial projection and the stated boundary conditions; it does not determine an absolute excitation energy without the mode norm. Replacing Y_{10} by Y_{00} gives zero direct dipole projection by angular orthogonality; other modes are not excluded.

Finite detuning. The conclusion need not require an exactly tuned external source. Replacing its carrier by $\omega + \rho + \delta$ gives

$$L(\delta) = I_A \int_0^1 \eta(t)e^{i\delta t}\,dt + I_B \int_0^1 \eta(t)e^{-i(2\rho+\delta)t}\,dt.$$

Since $\int_0^1 t\eta(t)\,dt = 1/4$,

$$|L(\delta) - L(0)| \le \tfrac{1}{4}(|I_A| + |I_B|)|\delta|.$$

The conservative bounds $|I_A| < 3/1000$, $|I_B| < 6/1000$ therefore imply $|L(\delta)| > 7/8000$ for $|\delta| \le 1/2$. This is a sufficient excitation bound for the prescribed local source, not a linewidth or a decay rate. Its duration is dimensionless, not a measured physical loading time. Neither calculation describes formation of the underlying Q-ball, self-sustained excitation, or coupling to a specified cosmological field.

Canceling one excitation is not stopping the Q-ball. The source-projection identity also supplies a simple linear control prediction. Let $d = e^{-i\rho t}c$, initially zero, and apply two identical physical pulses of the preceding family, with the same local rotating-frame waveform and strength ϵ, separated by $\Delta \ge 1$. Their total projected drive is

$$I_{\text{pair}} = \epsilon L(\delta)(1 + e^{-i\rho\Delta}). \tag{S19}$$

Consequently $\Delta = \pi/\rho > 1$ cancels the mode projection generated by the first pulse, even for a common local detuning δ. Both sideband terms in L are retained. This construction needs identical pulses in the rotating frame: in laboratory variables their sources obey $J_2(t) = e^{i\omega\Delta}J_1(t - \Delta)$. A sign-reversed second pulse at that same half-period instead doubles the rotating projected drive. For $|\delta| \le 1/2$, the preceding bound ensures a nonzero first-pulse projection. Relative to that projection, a timing error h leaves the remainder $2|\sin(\rho h/2)| \le \rho|h|$. These statements are analytic consequences of the fixed linear model. The following finite-time calculation tests the zero-detuning case only; it does not numerically establish the entire detuning or timing-error family. The common unknown modal norm cancels from the cancellation condition; extinguishing an arbitrary pre-existing excitation would require its complex amplitude and calibration. Other modes, continuum waves, radiation already emitted, and nonlinear backreaction are not canceled by this one condition. Neither zero modal projection nor zero linear radiation implies that the charged Q-ball itself has stopped rotating or disappeared.

A finite-time radial counterpulse test. We tested the two-pulse prediction by evolving the linearized complex radial $\ell = 1$ equation on the fixed nominal C2 background, with the conjugate physical channel retained. Four initially zero arms used no pulse, one pulse, two equal-sign local rotating-frame pulses separated by $\Delta = \pi/\rho$, and a sign-reversed second pulse at the same delay. The calculation used $R = 20$, $T = 6$, grid spacings $h = 0.04, 0.02$ and fourth-order Runge–Kutta time steps $h/8$, with forcing reported per unit linear drive amplitude. It is a radial linear PDE calculation, not a nonlinear three-dimensional evolution.

Both grids passed the predeclared projection-balance and comparison checks. The equal-sign pair reduced the final target projection below 2% of the single-pulse reference, while the sign-reversed pair agreed with twice the reference complex projection within 2%. These are numerical acceptance bounds, not interval-certified errors. The observed fine-grid ratio was approximately 3.61×10^{-5} (0.0036%); the coarse-grid ratio was approximately 1.44×10^{-4}. The balance includes the explicitly computed defect of the stored nominal mode on the spatial grid; the source-only discrepancy is reported separately. The global second-component defect norm, which includes the forced cutoff of the nominal mode at the finite boundary, actually increases on refinement; we do not claim global eigenfunction-error convergence. The short-time paired defect and projection checks concern the evolving, spatially confined response. Figure S4 also shows that the radial field norm need not vanish when the selected projection is suppressed. At $T = 6$ its fine-grid value was 0.0119332 for the equal-sign pair, compared with 0.00818567 for the single pulse. The mode-subtracted endpoint norm uses the nominal reconstruction $q - cA - \bar{c}B$ and is not an orthogonal energy decomposition. No total energy, charge removal, long-time stability or complete stillness of the charged background is inferred.

Changing the complete pulse shape. A subsequent fixed comparison used the same nominal linear radial $\ell = 1$ evolution, $R = 20$, $T = 6$ and $h = 0.04, 0.02$. It compared the annular drive above with a mode-adapted drive, using single pulses and equal-sign pairs separated by π/ρ. The new drive retains both sideband terms of a quintic turn-on construction and has a spatial cutoff equal to one for $r \le 4$ and zero for $r \ge 8$. The reference instead uses the annulus $1 < r < 2$ and a $\sin^2$ temporal window with one rotating frequency. Thus both spatial and temporal waveforms change; the comparison does not isolate a spatial cause.

Each single pulse was normalized to the same reduced-source norm

$$B_{\mathrm{src}} = \int\!\!\int |s(r,t)|^2 \,\mathrm{d}r\,\mathrm{d}t, \qquad B_{\mathrm{src}} \simeq 1.35281 \times 10^{-3} \quad (h = 0.02). \tag{S20}$$

Pairs have twice this budget. The integrated budgets agreed with their prescribed values within 6×10^{-9} relatively on both grids. This norm is neither mechanical work nor injected energy or charge, and equal norms do not imply equal actuator support or hardware cost. For the single pulse, the fine-grid diagnostic $|d(T)|^2/B_{\mathrm{src}}$ is about 32 times larger for the shaped drive (Table S1); this is not an energy-efficiency ratio or an optimality result.

Table S1: Fixed source-shaping comparison at $T = 6$, $h = 0.02$. Here $p = \partial_t q$, and "remainder" means subtraction of the nominal mode reconstruction. Field norms are $L^2(\mathrm{d}r)$ norms, not energies; the pair rows give the total remaining field and velocity norms.

Diagnostic	Annular drive	Shaped drive
Single-pulse $\lvert d\rvert^2/B_{\mathrm{src}}$	0.001214	0.03872
Single-pulse q remainder norm	0.007992	0.002596
Single-pulse p remainder norm	0.01502	0.004954
Pair $\lVert q\rVert$	0.01193	0.001055
Pair $\lVert p\rVert$	0.02153	0.001930

The shaped pair leaves approximately 8.8% and 9.0% of the reference pair's field and velocity norms, respectively; both remain nonzero. Nominal mode subtraction is not an orthogonal

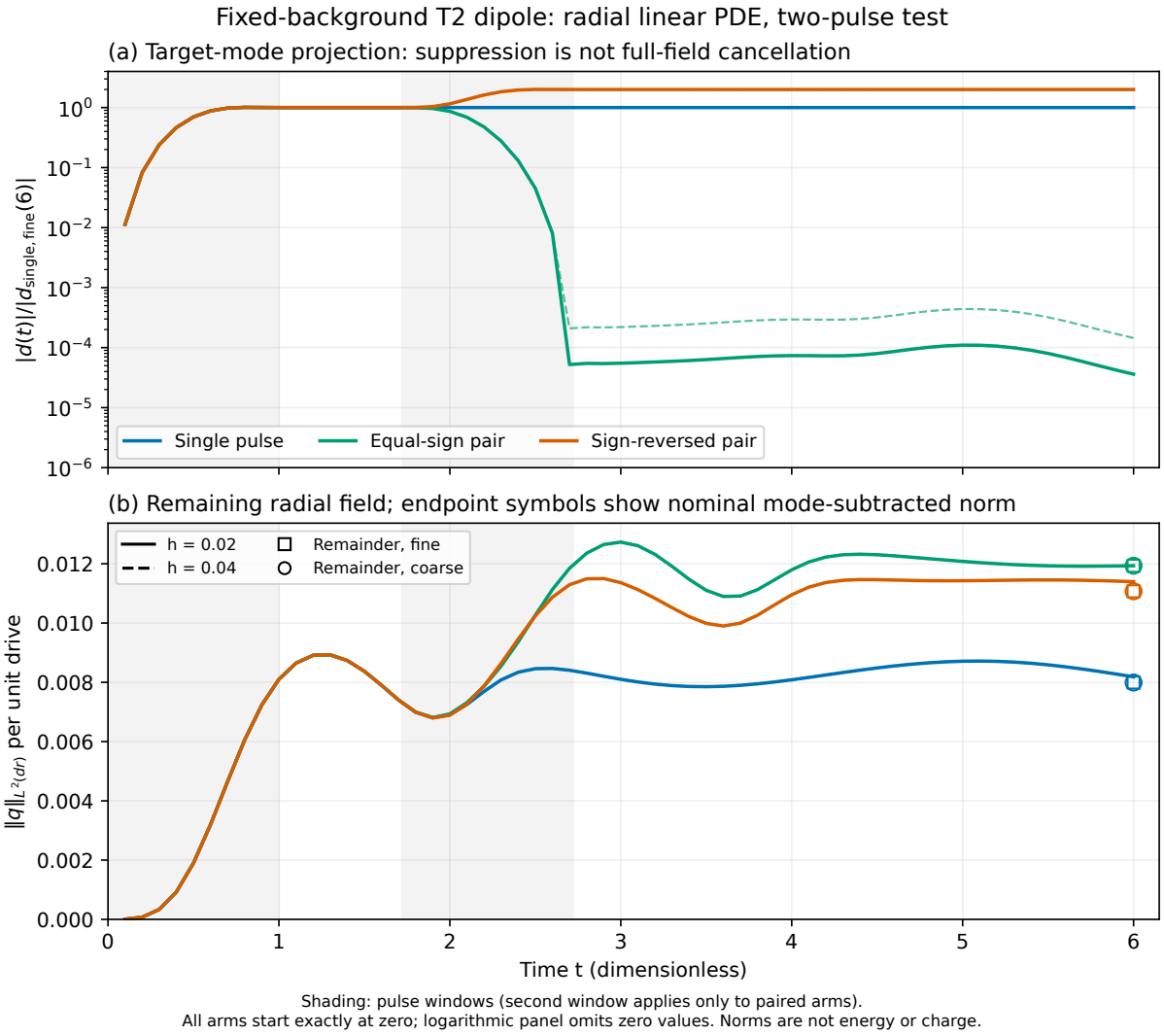


Figure S4: Stored radial linear-PDE histories for the C2 counterpulse test. Top: target projection magnitude, normalized by the fine-grid final single-pulse value. Bottom: reduced-field $L^2(dr)$ norms; endpoint symbols give norms after nominal mode subtraction. Colors identify pulse arms, solid/dashed lines the two grids. The exactly zero null arm is omitted from the logarithmic panel. Shading marks pulse support; only paired arms receive the second pulse. Lines connect recorded times without a new evolution. The laboratory second pulse includes the carrier-phase correction of the analytic construction. A small target projection coexists with a remaining field; these norms are neither radiated energy nor Noether charge.

energy partition. Both grids passed the fixed comparison and defect-balance checks, but the global second-component nominal eigenmode residual norm increases from approximately 0.529 to 1.495 on refinement. The successful paired-defect balance therefore does not establish global eigenfunction convergence. These finite-box, short-time results show reduced residual motion for the specified composite drive; they establish neither full stillness of the field nor removal of the charged background, nonlinear stability or a lifetime. This original two-source comparison did not separate changes of spatial and temporal design. The subsequently specified cross-comparison below addresses that limitation within a fixed actuator family.

A subsequent stored-profile audit identifies the origin of the growing second-component defect in this box. The exported reference mode has $v(20) = 0.004228503237$, whereas the evolution operator imposes a zero Dirichlet value. For the three-point stencil this discrepancy adds $b_j = \delta_{j,n-1} v(20)/h^2$ to the continued-ghost residual, giving $\|b\|_h^2 = |v(20)|^2/h^3$. Its squared norm is 0.2793787442 and 2.2350299533 on the two grids; the continued residual squared norms are 1.52×10^{-8} and 9.50×10^{-10}, with signed cross terms -9.74×10^{-10} and -4.84×10^{-10}. The boundary discrepancy therefore dominates this component's growing residual. It does not dominate the first component. This is an algebraic attribution, not a modification of the evolution boundary condition or a new eigenfunction calculation. The original global defect remains valid for the original operator. Two decreasing interior residuals neither certify the exported mode nor bound the integrated influence of the boundary discrepancy on the pulse response.

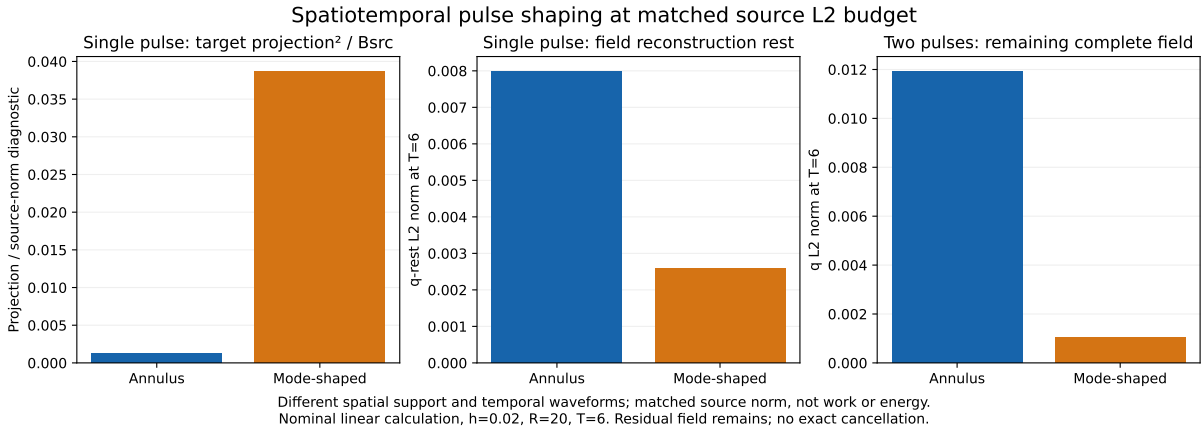


Figure S5: Endpoint diagnostics from the fixed spatiotemporal pulse comparison. Both spatial support and temporal waveform change. The common source norm is not energy or work. The shaped source increases the selected projection per source norm and reduces the residual response in this comparison; the complete field does not vanish. Bars show the stored fine-grid result, not a new evolution or a certified continuum limit.

A fixed factorial actuator comparison. The new comparison uses two spatial pairs, $A = (w, 0)$ and $B = (\zeta u, \zeta v)$, where w is the preceding annulus profile, ζ the cutoff defined above (one below $r = 4$ and zero above $r = 8$), and u, v the stored nominal sidebands. Two temporal pairs are prescribed on $0 < t < 1$:

$$T_0 = (\sin^2(\pi t) e^{i\rho t}, \sin^2(\pi t) e^{-i\rho t}), \tag{S21}$$

$$T_1 = ([g'' + 2i(\rho + \omega)g'] e^{i\rho t}, [g'' - 2i(\rho - \omega)g'] e^{-i\rho t}), \qquad g = 10t^3 - 15t^4 + 6t^5. \tag{S22}$$

Both temporal pairs vanish outside $0 < t < 1$; g is extended constantly as zero before the pulse and one afterward. Each complex source is the component sum $s_{ij} = \mathcal{N}_{ij}(R_{i1}T_{j1} + R_{i2}T_{j2})$, where $R_i = A$ or B. There is no conjugation in this sum; the second physical equation receives s_{ij}^*. The positive factor $\mathcal{N}_{ij}$ fixes the same single-pulse B_{src} *after* summation, retaining component interference. The two old corners A/T_0 and B/T_1 are implementation controls; A/T_1 and B/T_0

are the new crossed drives. Each uses both a single pulse and an equal-sign pair at the unchanged delay π/ρ. The nominal operator, $R = 20$, $T = 6$, $h = 0.04, 0.02$ and time step $h/8$ remain fixed. Both grids passed the predeclared numerical checks.

Table S2: Four fixed normalized actuator designs at $T = 6$, $h = 0.02$. The projection diagnostic is for a single pulse; the final two columns are total remaining pair norms, not mode-subtracted norms or energies.

Design	$\lvert d_{\mathrm{single}}\rvert^2/B_{\mathrm{src}}$	Pair $\lVert q\rVert$	Pair $\lVert p\rVert$
A/T_0	0.001214	0.01193	0.02153
A/T_1	0.0004100	0.006390	0.01426
B/T_0	0.1586	0.02024	0.005692
B/T_1	0.03872	0.001055	0.001930

Within this family, B/T_0 gives the largest single-pulse projection diagnostic, whereas B/T_1 leaves the smallest pair norms in both q and p. Strong projected excitation and small residual field are thus different objectives. The spatial-pair change $A \to B$ increases the pair q norm at T_0 but decreases it at T_1: its effect depends on the temporal partner. These are conditional design comparisons, not universal additive spatial and temporal effects. In particular A omits a spatial component, T_1 changes sideband weights and phase structure, and normalization depends on both factors. Equal source norm still implies neither equal work nor energy efficiency. All four pairs suppress their selected projection under the fixed criterion, while their full responses differ and remain nonzero. The nominal-mode defect and finite-box limitations above remain. That fixed-phase comparison did not test phase-error robustness; it established neither exact preparation, complete stillness nor a lifetime.

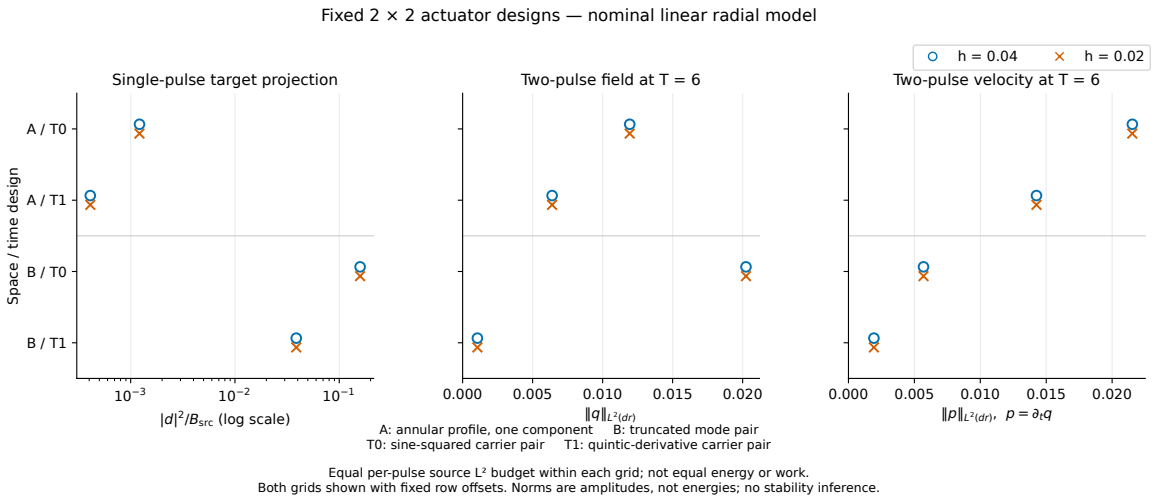


Figure S6: Stored endpoint diagnostics for the fixed 2×2 actuator family. The largest single-pulse projection per source norm occurs at B/T_0, while B/T_1 gives the smallest remaining pair field and velocity norms. The source is normalized after adding its two components; these comparisons are not energy efficiencies or a universal separation of independent spatial and temporal mechanisms. No additional evolution or phase-error test is represented by this figure.

A constant relative pulse phase. A further bounded test changes only the additional phase θ of the second pulse, keeping the delay and the carrier-phase correction fixed. The coupling to q^* makes the nominal Bogoliubov evolution real-linear, not complex-linear: the response to is need not equal iq. For each of the four fixed designs, one new delayed-is response therefore completes

the real basis. At the same endpoint,

$$q_\theta = q_{\text{single}} + \cos\theta\,(q_{\text{pair},0} - q_{\text{single}}) + \sin\theta\, q_{\text{delayed},90}. \tag{S23}$$

The same identity holds for p and the real-linear projection d. Four directly evolved $\theta = \pi/4$ pairs check this reconstruction: the largest reported relative discrepancy in q, p, d across both grids was 1.14×10^{-13}. This tests a known linear identity, not a new physical effect. Other constant phases below are evaluations of stored endfields, not additional time evolutions. The setting remains $R = 20$, $T = 6$, $h = 0.04, 0.02$.

Table S3: Selected constant-phase endpoints for B/T_1 at $T = 6$, $h = 0.02$. Norms are total remaining radial fields, not energy. The phase values were fixed before the endpoint analysis.

θ	$\|q\|$	$\|p\|$
0°	0.001055	0.001930
−5°	0.001062	0.002068
+5°	0.001423	0.002462
−15°	0.001986	0.003720
+15°	0.002562	0.004413
45°	0.006301	0.01125

These values need not be symmetric about zero phase. Cancellation of the selected coefficient and minimization of the full field norm are different conditions; no new minimum search was performed.

For a uniformly distributed phase that is *constant within each pulse pair*, write any endpoint field as $a + b\cos\theta + c\sin\theta$. Its root mean square norm is exactly

$$\sqrt{\mathbb{E}_\theta \|a + b\cos\theta + c\sin\theta\|^2} = \sqrt{\|a\|^2 + \tfrac{1}{2}(\|b\|^2 + \|c\|^2)}. \tag{S24}$$

The Gram-matrix evaluation was also checked against eight equally spaced phases. Table S4 keeps all four designs in the comparison. Although B/T_1 leaves smaller pair norms than A/T_1 at zero

Table S4: Uniform constant-phase RMS endpoint norms, $T = 6$, $h = 0.02$. These are square roots of mean squared norms, not mean norms.

Design	$\sqrt{\mathbb{E}_\theta \|q\|^2}$	$\sqrt{\mathbb{E}_\theta \|p\|^2}$
A/T_0	0.01128	0.02168
A/T_1	0.007123	0.01447
B/T_0	0.02552	0.04088
B/T_1	0.01106	0.01954

phase, its two RMS norms are larger in this ensemble. Its fixed-phase advantage is thus conditional on phase preparation. The comparison fixes source norm, not the previously excited target amplitude: A/T_1 has a much smaller single-pulse projection diagnostic (0.0004100 versus 0.03872 for B/T_1). Its lower RMS therefore does not establish a better storage or erasure protocol at equal preparation. An RMS ranking is not necessarily an ordering of mean norms. This ensemble is neither a time average nor a simulation of fluctuating environmental noise. The constant phase does not change the disjoint pulses' source budget $2B_{\text{src}}$, but that budget remains distinct from work or energy. The global nominal eigenmode residual limitation remains unchanged; neither exact BIC preparation, complete stillness nor nonlinear or long-time robustness follows.

Quadratic endpoint energy and the rotating generator. A further analysis uses the stored single- and pair-pulse endpoints, without repeating the time evolution. With $p = q_t$ and the same

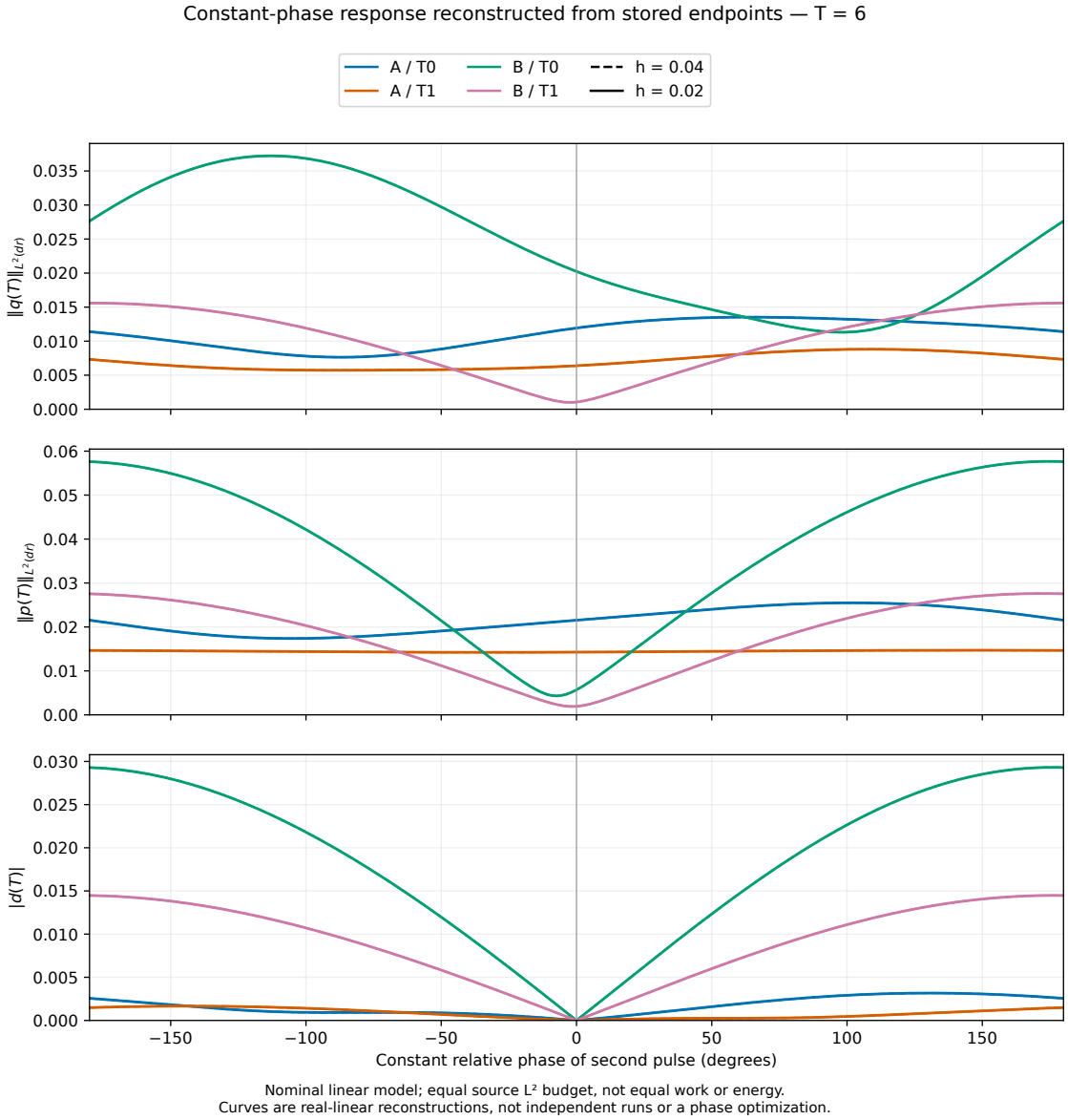


Figure S7: Constant-phase endpoint reconstruction for all four fixed actuator designs, from their stored real-linear response bases. The two grids compare nominal finite-box observables; plotted phase samples are not independent time evolutions. The additional phase is constant within a pulse pair, not a time-dependent noise source. Field norms and uniform-phase RMS values do not measure energy cost.

h-weighted inner product and Dirichlet operator as the evolution, define

$$H_2 = \|p\|_h^2 + \langle q, Kq\rangle_h + \mathrm{Re}\; h\sum_j C_j q_j^2, \tag{S25}$$

$$Q_2 = 2\,\mathrm{Im}\langle q, p\rangle_h + 2\omega\|q\|_h^2, \qquad E_2 = H_2 + \omega Q_2. \tag{S26}$$

Here $K = -\Delta_h + 2/r^2 + 1 - 4f^2 + (9/2)f^4 - \omega^2$ and $C = -2f^2 + 3f^4$. There is no factor $1/2$: H_2 is the coefficient of second order in $E - \omega Q$ for the stated action convention, using one physical complex field rather than counting its conjugate twice. This rotating quadratic generator need not be positive.

Table S5: Stored endpoint coefficients at $T = 6$, $h = 0.02$. All entries are in units of 10^{-4}, rounded to four significant digits. Within a fixed pulse count the designs have equal source-L2 budgets; a pair has twice the single-pulse budget.

Design	H_2 single	H_2 pair	E_2 single	E_2 pair
A/T_0	4.782	9.315	5.958	11.51
A/T_1	2.668	5.180	3.614	6.910
B/T_0	8.917	1.382	6.467	8.608
B/T_1	2.129	0.05203	1.201	0.03759

The distinction matters even for the direction of change: B/T_0 decreases H_2 while increasing E_2 from single to pair. Its Q_2 changes from approximately -2.821×10^{-4} to 8.319×10^{-4}. For B/T_1 both endpoint coefficients decrease; for the two A designs both increase. These directions also occur on $h = 0.04$, but two-grid differences are not validated error bounds.

Equal source-L2 budgets do not imply equal work, and the single-to-pair comparison itself is not a same-budget comparison. For the fixed semidiscrete model, $\dot{H}_2 = 2\,\mathrm{Re}\langle p, S\rangle_h$. Thus $H_2(T)$ at zero initial data predicts net rotating-generator work by an identity; that endpoint-only analysis did not measure an independent time-integrated power balance. The subsequent test below adds this observable. The quadratic laboratory source power additionally contains $2\omega\,\mathrm{Im}\langle q, S\rangle_h$, while $\dot{E}_2$ also contains the background-exchange term $2\omega\,\mathrm{Im}\; h\sum_j C_j q_j^2$. In particular, E_2 is a coefficient on the linear ansatz, not the complete laboratory energy change including a second-order background response. No positive or gross actuator work, energy efficiency, or full shutdown of the Q-ball is established by Table S5.

Time-integrated source work in the nominal linear model. A subsequent bounded run retains the same eight single/pair arms, $R = 20$, $T = 6$, $h = 0.04, 0.02$ and $\Delta t = h/8$, and adds power accumulators at the four Runge–Kutta stages. With the conventions of Eqs. (S25)–(S26), these integrate

$$P_H = 2\,\mathrm{Re}\langle p, S\rangle_h, \qquad J_Q = 2\,\mathrm{Im}\langle q, S\rangle_h, \qquad X_Q = 2\,\mathrm{Im}\; h\sum_j C_j q_j^2, \tag{S27}$$

$$P_{E,\mathrm{src}} = P_H + \omega J_Q, \qquad P_{E,\mathrm{ex}} = \omega X_Q. \tag{S28}$$

The independently evaluated state coefficients satisfy the numerical balances $\Delta H_2 = \int P_H\,\mathrm{d}t$, $\Delta Q_2 = \int (J_Q + X_Q)\,\mathrm{d}t$, and $\Delta E_2 = \int (P_{E,\mathrm{src}} + P_{E,\mathrm{ex}})\,\mathrm{d}t$. Unlike the preceding endpoint-only analysis, these integrals use the actual time-dependent fields and source at the integration stages. The exchange term remains active after the source ends.

The first pulse is identical in the single and pair arms; the first and second pulse windows are disjoint. Their trajectories therefore agree before the second pulse. Subtracting their independently integrated total source works gives the second pulse's net work on the already excited field:

$$W_{H,2} = W_{H,\mathrm{pair}} - W_{H,\mathrm{single}}, \qquad W_{E,\mathrm{src},2} = W_{E,\mathrm{src,pair}} - W_{E,\mathrm{src,single}}. \tag{S29}$$

This is not the work of an isolated second pulse applied to a fresh zero state, nor merely a relabeling of endpoint energy differences.

Table S6: Net second-pulse source work inferred from time-integrated pair-minus-single works at $h = 0.02$. Units are 10^{-4}, as in Table S5; entries are rounded to four significant digits. Negative values denote net extraction in the specified quadratic source-work convention, not efficiency of a physical actuator.

Design	$W_{H,2}$	$W_{E,\mathrm{src},2}$
A/T_0	4.533	4.207
A/T_1	2.512	3.602
B/T_0	−7.535	5.939
B/T_1	−2.077	−1.122

Thus the second B/T_1 pulse is a net source-work sink in both conventions. In contrast, B/T_0 removes rotating-generator work while its laboratory source-work coefficient is positive. The change in E_2 also includes the difference of integrated exchange terms; it is not identical to $W_{E,\mathrm{src},2}$. Positive and negative parts of the total spatially integrated power were separately accumulated, so a small net value need not mean little positive input or little extraction. This splitting applies after spatial summation, rather than to positive and negative local power densities.

Both grids pass the predeclared 10^{-6} normalized balance tolerance, and the new endfields agree with the stored reference endpoints within the fixed 10^{-8} single-response-normalized tolerance. The reported scaled two-grid differences of work contributions are below 4.4×10^{-4}, within the prescribed 0.03 consistency limit. These are discrete numerical checks, not rigorous continuum error bounds; the positive/negative power split also need not inherit a smooth fourth-order convergence rate. The original nominal-mode boundary defect is unchanged. Equal source-L2 budgets are still not equal work. Finally, the laboratory source-work coefficient and E_2 belong to the frozen linear ansatz: second-order background response, full nonlinear laboratory energy, actuator losses and conversion efficiency are not determined. No complete shutdown or long-time stability follows from this short-window balance.

Phase dependence of the second-pulse source work. For the same four source designs, we vary the constant phase θ of the delayed pulse while keeping its source norm fixed. Real linearity of the frozen evolution gives the quadratic source-work coefficient

$$W_{E,\mathrm{src},2}(\theta) = c_0 + c_1 \cos\theta + s_1 \sin\theta + c_2 \cos(2\theta) + s_2 \sin(2\theta). \tag{S30}$$

Writing the measured quadrature coefficients as (A, B, D, E, F), these are $(c_0, c_1, s_1, c_2, s_2) = ((D+E)/2, A, B, (D-E)/2, F/2)$. The five coefficients are integrated from the first-pulse response and two delayed quadrature responses, with a direct 45° two-pulse control, through $T = 3$ after both source windows end ($\Delta + 1 < 3$), using $h = 0.04, 0.02$ and $\Delta t = h/8$. This is not a renewed $T = 6$ energy history. Thus the plotted curves are postprocessing of measured coefficients, not thousands of independent evolution runs.

On both nominal grids, only B/T1 has a negative-work interval: approximately $-66.53° < \theta < 57.03°$. For the fine grid its second-pulse net source work is -1.1224×10^{-4} at 0° and -3.9278×10^{-5} in the direct 45° control. This broad nominal interval concerns signed net source work; it does not assert equally small residual fields throughout the interval. In particular, the previously reported residual field norms increase at 45° even though this work remains negative. All four uniform *constant-phase* means c_0 are positive; for B/T1 the fine-grid mean is 1.1334×10^{-4}. This phase ensemble is not a model of time-dependent phase noise.

Polynomial sign isolation treats the measured floating-point coefficients as exact rational dyads. Its exactness applies to those nominal polynomials, not to their discretization errors or to the

continuum or nonlinear field equations. The result concerns quadratic source work in the frozen linear ansatz, with background exchange distinguished separately; it is neither a full nonlinear laboratory-energy balance nor an actuator-efficiency claim.

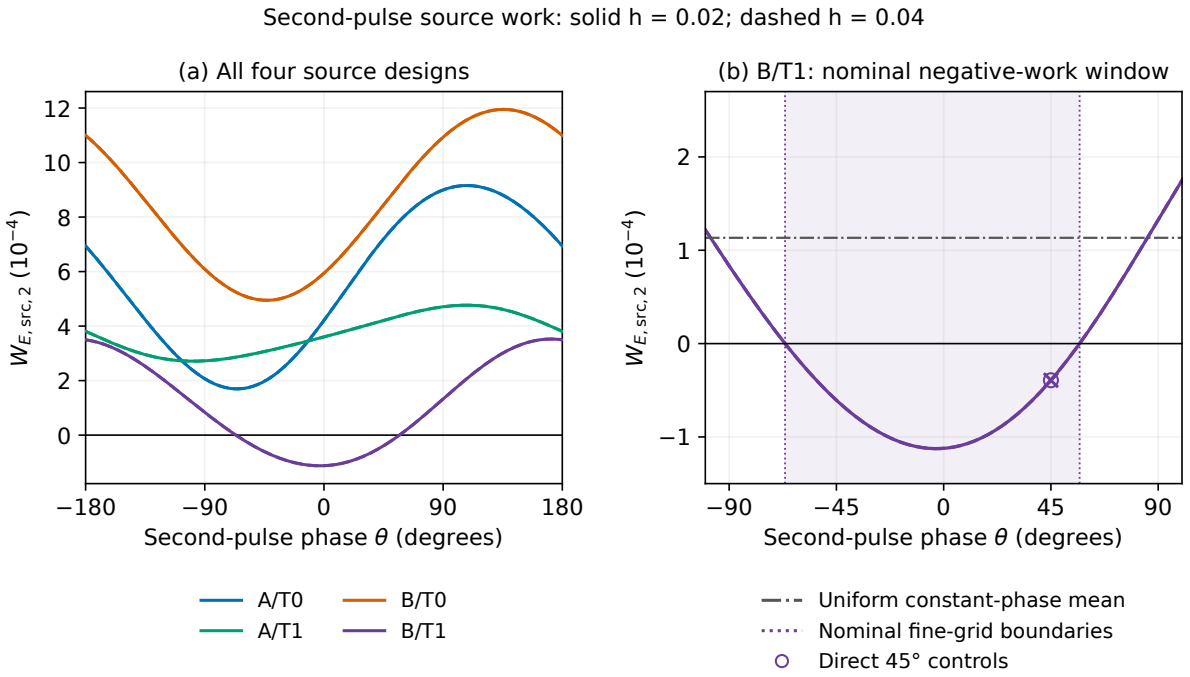


Figure S8: Second-pulse source work versus constant phase. Left: all four designs; solid curves use $h = 0.02$, dashed curves $h = 0.04$. Right: B/T1 with nominal fine-grid zero boundaries (dotted), its uniform constant-phase mean (dash-dotted), and direct 45° controls (symbols). Work is displayed in units of 10^{-4}. Shading denotes the nominal negative-work window, not certified physical tolerances.

One prescribed temporal phase drift. The constant-phase ensemble above does not determine the response to a phase varying during a pulse. As a separate finite comparison, we replaced the same delayed B/T1 source $S_d(t)$ by $S_{\delta,\theta_0}(t) = \exp[\mathrm{i}\theta_0 + \mathrm{i}\delta(t-\Delta)]S_d(t)$, with the single preselected value $\delta = 0.1$ against $\delta = 0$. The multiplier acts on the entire complex source; the original envelope, carrier construction and delay are unchanged. It preserves the source magnitude pointwise and its integrated L^2 budget. For each detuning we independently evolved the responses to $S_{\delta,0}$ and $\mathrm{i}S_{\delta,0}$ through $T = 3$. Real linearity gives the response at θ_0 as the cosine–sine combination of these two responses; the second is not assumed to be i times the first.

Averaging uniformly over the *constant initial* phase θ_0 eliminates both the mixed quadrature term and the work cross term with any deterministic first-pulse field. Consequently the mean second-pulse laboratory source work is one half of the sum of the two isolated quadrature works. No first-pulse evolution was repeated; this reduction applies to the mean, not to phase-specific two-pulse work. With $(h, \Delta t) = (0.04, 0.005)$ and $(0.02, 0.0025)$, both means are positive on both discretizations. On the finer grid the mean increases from 1.133373236×10^{-4} to 1.248632464×10^{-4}, a difference of $1.152592277 \times 10^{-5}$ (about 10.2%).

The increase exceeds the predeclared diagnostic comparison guard 7.94×10^{-11}, which combines the change between discretizations, unshifted-work reproduction errors and energy-balance residuals. This guard is not a certified error bound; space and time resolution change together. The result establishes sensitivity of this nominal mean source work to one prescribed phase drift at equal source norm. It does not give a phase-noise or thermal model, an efficiency gain, or a claim about all detunings. The frozen-profile approximation, artificial $R = 20$ boundary and omitted nonlinear backreaction remain unchanged.

Local direction and one mirror detuning. A follow-up fixed $d = 0.1$ and added only the mirror source at $-d$ and the parameter tangent at zero. Writing $z = \partial_\delta q|_0$, the exact differentiated finite-grid equation has the same real-linear operator and forcing $\mathrm{i}(t-\Delta)S_d$. Its numerical integration retains both derivatives of the work integrand, from the response and from the source, for each independently evolved quadrature. The zero-detuning basis trajectories are needed for these products; the first pulse is not repeated, and the $+d$ result is reused. For the uniform initial-phase mean $m(\delta)$, the finer grid gives $m'(0) \simeq 1.1006065 \times 10^{-4}$ in model units. Both grids resolve a positive slope with the prescribed empirical guard. The three fine-grid means remain positive:

δ	Mean laboratory source work $m(\delta)$
-0.1	1.0285198×10^{-4}
0	1.1333732×10^{-4}
$+0.1$	1.2486325×10^{-4}

Define the finite odd and even changes by $O_d = [m(d) - m(-d)]/2$ and $E_d = [m(d) + m(-d) - 2m(0)]/2$. Their fine-grid values are approximately 1.1005634×10^{-5} and 5.20289×10^{-7}; the residual $R_d = O_d - dm'(0) \simeq -4.31 \times 10^{-10}$ is small but not an exact identity or a certified Taylor remainder. The resolved positive odd change agrees with the local direction. This is a directional response of the fixed nominal actuator, not an identification of rotation as its cause: a free complex accelerator $q_{tt} = \exp[\mathrm{i}(\kappa+\delta)t]$ already admits a nonzero detuning slope. More generally, uniform initial-phase averaging selects the normal causal response, whose work derivative is a lag-weighted imaginary response kernel; the carrier and field response enter it jointly. The two-grid guards are diagnostic, and the inherited profile, boundary and linearization limitations persist. No thermal, new-memory-mediator or efficiency conclusion follows.

Prescribed finite phase coherence. A fluctuating drive phase can be specified without identifying it with thermal noise. Let the finite-duration source be $S_\theta(t) = e^{i\theta(t)}S_d(t)$, where the initial phase is uniform and independent and $\theta(t) - \theta(\Delta) = \sqrt{2D}\, B_{t-\Delta}$ during the pulse, with B a

standard real Wiener process independent of that initial phase. Then $\mathbb{E}e^{i[\theta(t)-\theta(s)]} = e^{-D|t-s|}$ and the anomalous source correlation vanishes. Every realization retains the source's L^2 budget. This does not fix the work delivered to the field.

For the fixed finite real-linear response used above, let $m(\delta)$ denote the phase-averaged source work when the *entire* prescribed source is multiplied by $e^{i\delta(t-\Delta)}$. Its phase-diffusion mean obeys

$$m_D = \int_{-\infty}^{\infty} \frac{D}{\pi(D^2+\delta^2)}\, m(\delta)\, d\delta, \qquad D > 0.$$

This follows from the characteristic function of the Cauchy distribution and the quadratic response kernel. At finite time the finite-dimensional propagator is bounded; compact source support and finite L^2 norm give absolute integrability and justify interchanging the integrals. The same weighting convolves the finite-pulse spectral power and preserves its integrated weight. Since the weight is even, the odd detuning response cancels. The positive local slope found above therefore does not determine whether phase diffusion increases or decreases mean work. Nor does a single symmetric detuning pair settle this question.

For computation of this mean work, the lag factor $e^{-D(t-s)}$ can instead be included in the auxiliary generator $A - DI$, evaluated for two genuine source quadratures. With state coordinates (q, p), both equations acquire $-D$ terms; the work output remains $p + i\omega q$, although now $\dot{q} = p - Dq$. These auxiliary states are not stochastic field trajectories, and their endpoint energies are not ensemble energies. This reduction concerns the specified work observable and does not insert physical friction into the original field equations.

A Cauchy-distributed constant detuning and a Wiener phase process share the relevant two-point statistics but not their trajectory laws. Their quadratic response means agree; the auxiliary damped endpoint is a different object. Ideal Wiener phase also has no finite ultraviolet cutoff. Thus this identity is a finite-model interpretation, not a continuum certificate, a temperature assignment, or a derivation of an environmental bath. Physical bandwidth and coupling require separate specification before experimental use.

One finite-coherence comparison. We fixed $D = 1$, corresponding to one pulse duration for the exponential phase-correlation time, and compared it with $D = 0$. The two genuine quadratures of each auxiliary response were evolved on the same two discretizations as the detuning test. The finer grid gives mean source works $m_0 = 1.133373236 \times 10^{-4}$ and $m_1 = 2.173314375 \times 10^{-4}$. Their difference, $1.039941139 \times 10^{-4}$, is positive on both grids and exceeds the predeclared diagnostic guard 6.63×10^{-9}. The undiffused adapter reproduces the earlier work values, and the modified auxiliary balances include $-2D$ times each quadratic form. Free and restoring-oscillator controls test both the work output and the placement of damping in both auxiliary coordinates.

This is an increase in mean work supplied by the prescribed external source at fixed source L^2 budget. It is not a gain in energetic efficiency or a demonstration of self-sustained oscillations. The guard combines grid differences and implementation residuals; it is not a certified error bound, and space and time resolution change together. The nominal profile, finite boundary and linear-response limitations persist. Only one nonzero diffusion constant is compared: no noise optimum, temperature, nonlinear formation or lifetime conclusion is established. The same source-output reduction with output p gives an additional observable. In the undamped physical finite model, the rotating quadratic generator satisfies $\dot{H} = 2\,\mathrm{Re}\langle p, S_\theta\rangle$. For zero initial perturbation, its ensemble endpoint mean therefore equals the half-sum of the two auxiliary integrals of this *original* source rate. It does not equal the auxiliary endpoint H. Extracting these already stored integrals gives, on the finer grid, $\mathbb{E}H(T) = 2.09020168 \times 10^{-4}$ for $D = 0$ and $2.93069596 \times 10^{-4}$ for $D = 1$. Both grids show this approximately 40% increase; this is a descriptive follow-up without a new certified comparison bound or field integration.

This rotating quadratic generator need not be positive, and its global value does not measure energy localized near the Q-ball. Full laboratory endpoint energy additionally requires the physical ensemble charge, including the anomalous state-exchange contribution. The latter is not given by

the exchange evaluated on the damped auxiliary states. Thus neither the source-work increase nor this rotating-generator mean establishes laboratory energy retention or improved confinement.

Correlated instantaneous frequency. A separate prescribed drive model replaces Wiener phase increments by $\theta(t) = \theta_0 + \int_\Delta^t \xi(s)\,ds$, with an independent uniform initial angle and stationary zero-mean Ornstein–Uhlenbeck frequency $\mathbb{E}\xi(t)\xi(s) = (D/\tau_c)e^{-|t-s|/\tau_c}$. Its phase correlation at lag $u \geq 0$ is

$$C(u) = \exp\{-D[u - \tau_c(1 - e^{-u/\tau_c})]\}.$$

This retains the pathwise source norm and fixes the long-time phase diffusion constant D; the instantaneous frequency variance is D/τ_c. The Wiener kernel is recovered as $\tau_c \to 0$ at fixed D. Finite τ_c removes its linear short-lag cusp, but does not impose a hard bandwidth cutoff or define a thermal bath. Pointwise larger phase correlation need not give larger work against an oscillatory response.

Small correlation-time limit. For a fixed finite spatial operator and fixed $D \geq 0$, let $\tau = \tau_c$ and write the exact mean work as $m_\tau = \int_0^T W(u)C_\tau(u)\,du$, with real lag kernel W and $m_D = \int_0^T W(u)e^{-Du}\,du$. If $|W(u)| \leq M$, then

$$|m_\tau - e^{D\tau}m_D| \leq \frac{e^{D\tau}DM\tau^2}{1 + D\tau}, \qquad m'_+(0) = Dm_D.$$

If W is continuous at zero, the present source-work observable has $W(0) = 2B_{\mathrm{src}}$, where $B_{\mathrm{src}} = \int_0^T \|S(t)\|^2 dt$ is the exact nominal source budget. Consequently, $m_\tau = m_D + Dm_D\tau + (D^2m_D/2 - 2DB_{\mathrm{src}})\tau^2 + o(\tau^2)$. These local asymptotics neither determine the comparison at $\tau = 0.1$ nor imply global monotonicity, a thermal interpretation, or a continuum error bound.

For the fixed choice $D = 1$, $\tau_c = 1/10$, put $a = 1/10$. The mean source work has the signed representation $m_{\mathrm{OU}} = e^a \sum_{n\geq 0}(-a)^n m_{1+10n}/n!$, where m_λ is the two-quadrature work obtained from the auxiliary generator $A - \lambda I$. This is not a positive mixture of dissipative states. The selected truncation retains $n = 0, \ldots, 8$. If $B \geq 2\int_{t>s} |K_{\mathrm{work}}(t,s)|\,dt\,ds$, its analytical work-tail bound is

$$\epsilon_{\mathrm{tail}} \leq B\,\frac{10}{9}\,\frac{(1/10)^9}{9!}\,\frac{1}{1 - 1/100}.$$

The finite-model bound uses a positive reference energy norm, a coefficient-envelope check and an exact-rational source-norm bound for frozen nominal binary coefficients. This bounds the omitted series, not floating exponential evaluation, signed-sum roundoff or RK integration. Those errors and the combined space/time sensitivity require separate assessment; auxiliary endpoint energies are not ensemble energies. The first finite-correlation implementation passed its scalar controls but failed the preselected modified quadratic-balance tolerance on the coarser field grid. That implementation therefore yielded no accepted finite-correlation field result, and its finer grid was not run. The analytical series-tail estimate did not repair this numerical acceptance failure, and the original outcome and thresholds remain unchanged. A separate calculation with a different integrator and a prospectively specified temporal-sensitivity comparison is reported below.

A separate finite-correlation pulse comparison. A separately specified two-stage Gauss calculation retained the prescribed source, its normalization, $D = 1$, $\tau_c = 0.1$ and the signed nine-term expansion. It split integration steps at the two known source-window joins and used both genuine quadratures for every auxiliary rate, plus a zero-source control. Three fixed integrations compared $(h, \Delta t) = (0.04, 0.005)$, $(0.04, 0.0025)$ and $(0.02, 0.0025)$ through $T = 3$ at $R = 20$. All 108 recorded acceptance checks passed; this comparison was made against the previously stored Wiener-source work on the corresponding spatial grid, not against an auxiliary endpoint energy or a newly repeated Wiener run.

On the finer spatial grid, the truncated OU mean source work was $2.1932622828 \times 10^{-4}$, compared with $2.1733143747 \times 10^{-4}$ for the stored Wiener result. The difference, $1.9947908110 \times 10^{-6}$ (approximately 0.918%), exceeded the predeclared diagnostic guard $1.2658579115 \times 10^{-8}$ and had the same positive sign on the coarser grid. The guard includes the analytical series-tail allowance and measured temporal, spatial, adapter and accounting contributions; it is not a certified total-error bound. The fixed coarse-grid time comparison does not directly measure fine-grid temporal error. This is a bounded increase in mean source work for one prescribed drive in the frozen finite-domain model, not evidence of retained energy, heating, formation, improved efficiency or experimental realizability.

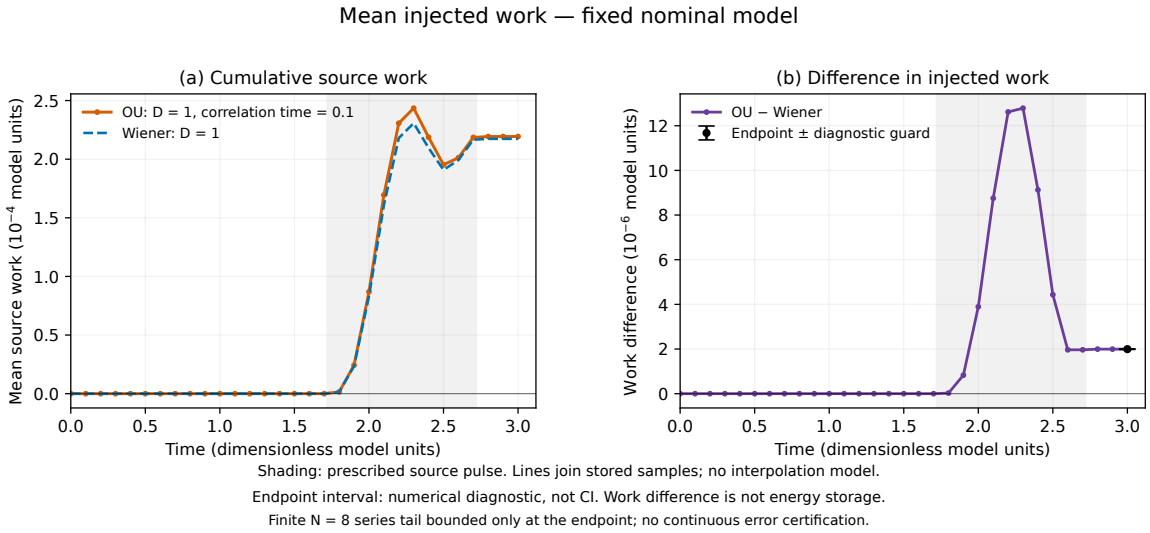


Figure S9: Cumulative mean source work for prescribed OU and Wiener phase drives on the $h = 0.02$ grid (left), and their difference (right). The OU curve is the signed nine-term mean of the two genuine quadrature works per auxiliary rate; the Wiener curve uses the previously stored calculation. Both use integrated original source work, not auxiliary endpoint or ensemble energy. Straight lines join 31 stored times, with no additional field integration for the figure. The endpoint marker spans the predeclared diagnostic guard; it is neither a confidence interval nor a certified total-error bar. All quantities are in dimensionless model units.

One exterior-domain sensitivity check. A separate calculation at fixed $h = 0.04$, $\Delta t = 0.0025$ and $T = 3$ extended $R = 20$ to $R = 30$, retaining byte-identical source values and interior coefficients and comparing with the stored $R = 20$ calculation. Across the 31 shared sample times, the absolute-weighted majorant for the change in the same-method OU-minus-$n = 0$ source-work contrast was 1.14×10^{-19}, below the predefined diagnostic threshold 1.27×10^{-9}. This is agreement at floating-point roundoff scale, not absolute accuracy at 10^{-19}; common spatial, temporal and profile errors can remain. The check does not establish an infinite-domain limit, long-time behavior or control between samples, and neither replaces the earlier Wiener baseline nor reduces the original comparison guard.

Signed contributions from source-time separations. A separate response-kernel calculation uses the fixed $h = 0.04$, $R = 20$ spatial operator and both genuine quadratures of each source profile. After averaging the uniform initial phase, its mean source work is $m[C] = \int_0^1 W(u)C(u)\,du$, where u is the separation of two source times, not the observation time. The kernel can have either sign, with $W(0) = 2B_{\mathrm{src}}$. On the finest of three lag meshes (400, 800, 1600 subdivisions), with independent overlap quadratures of orders 16 and 32, the four prescribed quarter-interval

contributions to $\int_0^1 W(u)[C_{\mathrm{OU}}(u) - e^{-u}]\,du$ are, in units of 10^{-6},

$$(+23.13074873,\ -9.08245885,\ -11.01641674,\ -1.03681136).$$

Their signed sum is $1.99506177879 \times 10^{-6}$, with summed heuristic refinement envelopes 6.98×10^{-14}; these are not a certified error bound or confidence interval. The shortest-lag contribution exceeds the three negative later-lag contributions (Fig. S10). Closure to the stored same-grid Gauss OU and Wiener-rate works passes the predeclared diagnostic allowances. The separate shift of the Gauss Wiener-rate work relative to the old RK Wiener baseline is -8.47524×10^{-10}; it is not assigned to any lag bin. This calculation is not the $h = 0.02$ comparison above and does not exactly decompose its mixed-integrator result. A lag bin integrates pairs of source times, not instantaneous positive or negative power. All separations are at most one model time unit, below the nominal period $2\pi/\rho \simeq 3.44$; the response involves the full generator, not an isolated BIC or a long ringdown. No stored-energy, formation or stabilization conclusion follows.

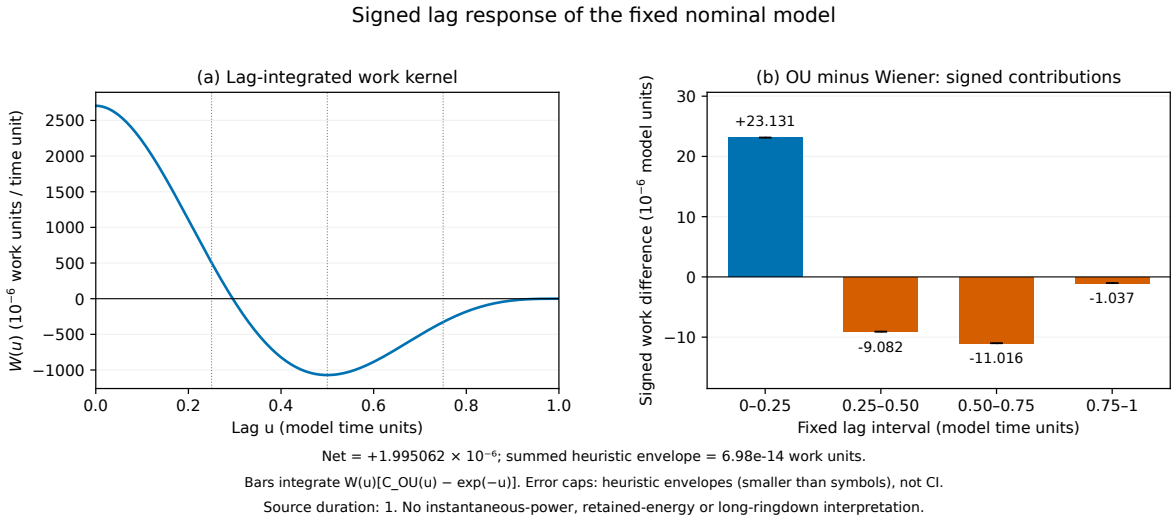


Figure S10: Stored finest-grid signed lag kernel $W(u)$ (left) and four fixed integrals of $W(u)[C_{\mathrm{OU}}(u) - e^{-u}]$ (right), for the frozen $h = 0.04$ model. Lines connect the stored lag nodes without smoothing. The caps and reported net envelope are heuristic numerical sensitivity measures, too small to resolve at this scale, not confidence intervals or certified errors; no continuous kernel error band is asserted. The positive net source-work difference retains all three negative contributions. These are source-time-pair contributions, not a time history of instantaneous power or retained energy.

Scalar radiation pressure has been studied in a related one-dimensional Q-ball model [3]; its rates are not imported into our three-dimensional problem.

Several Q-balls of the same nonlinear field can influence one another, but a sum of isolated stationary profiles is not automatically a stationary multi-centre solution. Separation, relative phase and incident frequencies must be specified. We have not derived a self-consistent environmental noise spectrum, an interaction with every particle species, or a mechanism that maintains these excitations indefinitely.

If an additional dynamical mediator is introduced, its spatially nonlocal static response must be distinguished from temporal memory. Matching a homogeneous effective potential does not generally match the nonlocal static interaction. Such extensions are not included in Eq. (1).

An additional open channel also changes the localization condition. For a real-frequency mode with independent propagating asymptotic channels, every propagating tail must vanish separately. A cancellation of the original outgoing amplitude therefore does not establish an embedded bound state of the extended system. Nor does a zero of a leading-order source overlap establish this:

the coupled background and mode can change at higher order. These are conditions for a future coupled calculation, not results for an extension of the model studied here.

S5 Exploratory concentration and constrained transport

A distinct formation hypothesis is that an initially disordered scalar field amplifies a band of spatial wavelengths and subsequently fragments. Condensate fragmentation and a scale selected by growing perturbations are established questions in other Q-ball models [6]. They do not establish formation in our initial-data family or select the BIC frequencies. A future comparison should vary narrow-band and broadband initial spectra at matched total energy and charge, measure growth and outgoing flux, and distinguish transient concentrations from persistent localized charged objects. A connected high-density region in a periodic box should also be checked for closed paths with nonzero winding around the box. A region with such a path is not an isolated density island at that threshold. Absence of winding alone would still not establish vacuum isolation, stationarity or stability; threshold and resolution dependence must be checked. Total charge conservation forbids production of net charge from neutral data; opposite charges or a compensating background would have to be accounted for. No universal integer multiple of a fundamental wavelength is derived here.

Even persistence alone can be biased by the preparation. For the potential in Eq. (1), put $S = |\phi|^2$, $\Pi = \partial_t \phi$, $Q = 2\,\mathrm{Im}\int \phi^* \Pi \,\mathrm{d}^3x$ and $q = |Q| > 0$. For finite total energy $E = \int (|\Pi|^2 + |\nabla\phi|^2 + \mathcal{U}(S))\mathrm{d}^3x$, the elementary identity

$$E - q = \int \left[|\Pi - \mathrm{i}\,\mathrm{sgn}(Q)\phi|^2 + |\nabla\phi|^2 - S^2 + \tfrac{1}{2}S^3 \right] \mathrm{d}^3x$$

implies $\int S^2 \mathrm{d}^3x \geq q - E$ when $E < q$. Since $\mathcal{U}(S) \geq S/2$, it also gives $\operatorname{ess\,sup} S \geq (q-E)/(2E) > 0$. Thus conserved whole-system energy and charge already preclude uniform dilution to arbitrarily small amplitude for such initial data. This elementary bound is not a formation result: it specifies neither a fixed spatial core nor a single stationary Q-ball. Charge transport, fragmentation and continued oscillation remain possible. A formation test must therefore resolve localization, profile and phase dynamics rather than count survival alone. The bound concerns the closed system, not an unbalanced subvolume or a dissipative boundary model.

Preparing a matched-energy and matched-charge width comparison. A restricted, analytically tractable preparation uses $\phi = A\exp[-r^2/(2R^2)]$ and $\Pi = \mathrm{i}Q\phi/(2I)$, where $I = \int |\phi|^2 \mathrm{d}^d x = A^2 \pi^{d/2} R^d$ and $Q \neq 0$. For the same potential in d spatial dimensions its energy is

$$E_d(I, R, Q) = \frac{Q^2}{4I} + \left(1 + \frac{d}{2R^2}\right) I - \frac{I^2}{(2\pi)^{d/2} R^d} + \frac{I^3}{2\,3^{d/2}\pi^d R^{2d}}.$$

The chosen momentum minimizes kinetic energy at fixed field and charge; it does not make the Gaussian a stationary solution. At fixed R, Q this expression has one positive-norm minimum $E_{\min}(R, Q)$. Two prescribed widths R_1, R_2 can therefore share (E_0, Q) within this preparation family precisely when $E_0 \geq \max_j E_{\min}(R_j, Q)$. Above each minimum there are two positive solutions for I; a comparison must fix its branch in advance, for example the smaller-I solution. Actual discrete energies and charges must be checked separately. The factors of d describe spatial dimension, not additional field components. This width comparison does not replace the disordered-spectrum test above, and for $E_0 < |Q|$ both admissible preparations already obey the non-dilution bound. No formation run using this prescription is reported here.

A separate finite-time radial relaxation study. A subsequent project study of the same potential evolved four prescribed spherically symmetric Gaussian initial states to $T = 1000$, with an absorbing exterior. Its predeclared diagnostic compared the magnitude of the charge inside $r < 60$ with the stationary-family charge at the frequency estimated from the central phase, allowing a relative difference below 10%. This tests charge–frequency consistency, not equality of the evolving field to a stationary solution. The matched-width state passed at $T = 500$

and the broader state at $T = 1000$. The narrower state failed and retained about 2% of its initial charge inside the measurement region. A larger state remained strongly pulsating and also failed this diagnostic. Consequently, the combined formation criterion fixed before the runs was not met. The successful initial states already had whole-system $E/|Q| < 1$; they do not demonstrate capture of initially unbound radiation. Two jointly refined spatial and temporal resolutions gave the same classifications, but absorber reflection was not measured. These reported finite-time radial observations neither establish exact relaxation nor test nonspherical stability; a charge-retention plateau alone is not a binding proof. The original plan and result are identified in the reproducibility inventory. They have received a document-scope review, not an independent reproduction of the trajectories.

One angular direction during chirped concentration. To examine a freedom excluded by radial evolution, we evolved one prescribed quadrupolar tangent on a time-dependent inward chirp. This is an exploratory formation diagnostic on a different background from the stationary BIC studied above. In the units of Eq. (1), the radial initial data are $F(r) = (3/5)^{3/2}\exp(-r^2/50)\exp(\mathrm{i}\beta r^2)$, $\phi_b(r,0) = F(r)$ and $\partial_t\phi_b(r,0) = 0.8\mathrm{i}F(r)$, with $\beta = 0.015$. In this paragraph and the subsequent chirp-compensation calculation, β denotes the chirp parameter; the sextic potential coefficient remains fixed at $1/2$. Their whole-space analytic values are $Q \simeq 240.55177$ and $E/Q \simeq 1.038669965 > 1$; hence the preceding $E < |Q|$ non-dilution argument does not force a remnant. The calculation uses a radial Dirichlet box $R = 80$ and ends at $T = 32$.

For $\delta\phi = \eta(r,t)Y_{20}$, with real unit-normalized Y_{20}, put $w = r\eta$ and $S_b = |\phi_b|^2$. The laboratory-frame tangent equation is

$$w_{tt} = w_{rr} - \frac{6w}{r^2} - (1 - 4S_b + \tfrac{9}{2}S_b^2)w - (-2 + 3S_b)\phi_b^2\overline{w}. \tag{S31}$$

We fix $w(r,0) = r^3(-1/25 + 2\mathrm{i}\beta)F(r)$ and $w_t(r,0) = 0.8\mathrm{i}w(r,0)$, the regular $O(r^3)$ quadrupolar deformation. The nonlinear radial background is advanced simultaneously, without angular backreaction. Its stored diagnostics reproduce the earlier chirp evolution; this replay is a control, not independent formation evidence.

Centered second differences and RK4 are compared at the fixed choices

$$(h, \Delta t) = (0.1, 0.02), (0.05, 0.02), (0.1, 0.01).$$

Origin polynomial, independent free-evolution and conjugate-coupling controls precede the calculation. Our positive reduced diagnostic is

$$N_h^2 = h\sum_{j=1}^{J-1}\left[|w_{t,j}|^2 + \left(1 + \frac{6}{r_j^2}\right)|w_j|^2\right] + \frac{1}{h}\sum_{j=0}^{J-1}|w_{j+1} - w_j|^2, \qquad r_J = 80. \tag{S32}$$

It is not a conserved perturbation energy. Positive core, exterior and collar contributions use fixed intervals $[0,6]$, $[6,76]$ and $[76,80]$, with shared-node half weights and complete edge contributions.

On the finer spatial grid, the largest *sampled* $N_h(t)/N_h(0)$ is 1.356829 at $t = 21.6$; its value at $T = 32$ is 1.139233. Samples are separated by 0.8, so this is not a continuous-time supremum. The deformation concentrates in the core and subsequently redistributes outward (Fig. S11). Common-grid endpoint differences, measured in Eq. (S32) and divided by the base-grid initial norm, are 0.0062849 for spatial refinement and 4.0021×10^{-7} for timestep halving. These are sensitivity measures, not certified error bounds. The sampled collar contribution satisfies $N_{\mathrm{collar}}^2(t)/N_h^2(0) \le 10^{-6}$, the prescribed contamination threshold. This single direction shows transient amplification, not a stability classification: the background remains nonstationary, and other radial shapes, angular sectors, finite amplitudes and later times remain untested.

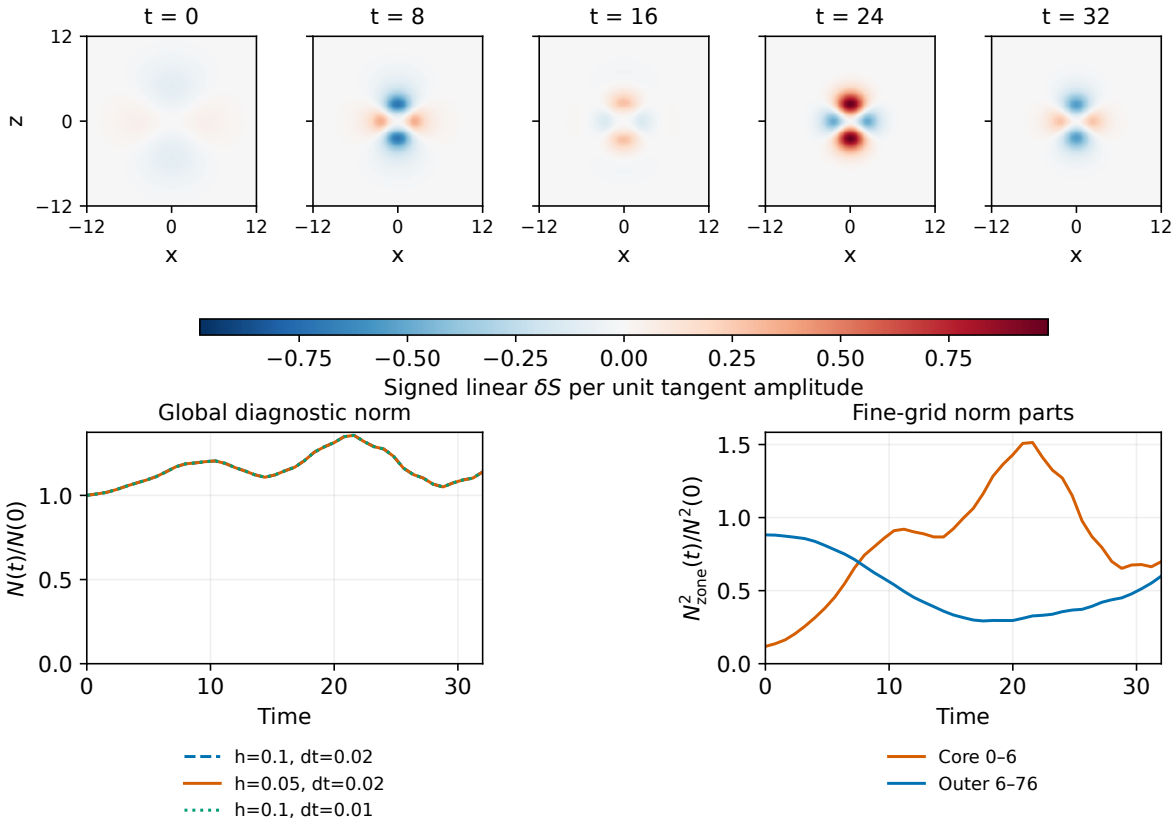


Figure S11: Angular response of the inward chirp. Top: the actual signed linear density variation $\delta S = 2\,\mathrm{Re}(\overline{\phi_b}\eta)Y_{20}$ from fine-grid snapshots at $t = 0, 8, 16, 24, 32$, in the fixed xz window $[-12, 12]^2$. Radial nodal values are interpolated for rendering; the common symmetric colour scale is per unit tangent amplitude. Bottom: global norm for all three discretizations, and fine-grid core/exterior squared-norm contributions divided by the *global initial* $N_h^2(0)$. These are linear tangent data, not a finite-amplitude nonlinear three-dimensional evolution or a physical energy partition.

Leading angular feedback on core-charge transport. A separate second-variation calculation follows $\phi_\varepsilon = B + \varepsilon\eta Y_{20} + \varepsilon^2\Xi + O(\varepsilon^3)$ and evolves the forced spherical mean $\zeta = \langle\Xi\rangle_\Omega$. Here Ξ is half the second derivative, and ζ is an angular mean, not the coefficient of normalized Y_{00}. Only this mean is needed alongside the first tangent for the angle-integrated charge through second order. The initial family is the volume-preserving ellipsoidal deformation $r_\varepsilon^2 = e^{-2c\varepsilon}(x^2+y^2) + e^{4c\varepsilon}z^2$, $c = \sqrt{5/(16\pi)}$, with an explicitly differentiated finite-grid normalization fixing total charge. For Q_2, the coefficient of ε^2, define

$$b = \frac{Q_{2,\mathrm{core}}(32) - Q_{2,\mathrm{core}}(0)}{Q_0}$$

in the same fixed core $r < 6$, box $R = 80$ and time window. The three $(h, \Delta t) = (0.1, 0.02), (0.05, 0.02), (0.1, 0.01)$ calculations give $b = 0.08737135$, 0.08736583, 0.08737135, respectively; the spatially refined value exceeds the predeclared heuristic guard 2.864×10^{-5}. This guard combines measured spatial, temporal and charge-balance sensitivities with a fixed floor; it is not a certified error bound or confidence interval. Stage-integrated physical charge flux checks the core-charge difference, rather than defining it from the endpoint.

The interpretation requires the initial redistribution (Fig. S12). On the finer spatial grid, $Q_{2,\mathrm{core}}(0)/Q_0 = -0.10587389$, whereas $Q_{2,\mathrm{core}}(32)/Q_0 = -0.01850805$. The radial reference itself has positive net core inflow over this window: its core fraction increases from 0.58949606 to 0.65267983. The positive second-order net-transport coefficient recovers much of an initial core-charge deficit but leaves the final correction negative; it does not show more final charge retained than in the radial reference. The sampled response is nonmonotonic, with a transient positive core correction before the negative final value; the decision uses the fixed endpoint $T = 32$, not a selected intermediate maximum. The preparation fixes charge, not energy: $E(\varepsilon, 0) = E_0 + \varepsilon^2 E_2 + O(\varepsilon^3)$ with $E_2/E_0 = 0.0224455$ on that grid. These are second-order Taylor coefficients at zero deformation, not a finite-amplitude nonlinear three-dimensional evolution, a binding comparison at equal energy, or evidence of stationary Q-ball formation. As a subsequent analytical interpretation, the untruncated continuum family has $E_{\mathrm{grad}}(\varepsilon) = E_{\mathrm{grad},0}[2e^{-2c\varepsilon} + e^{4c\varepsilon}]/3$, while its kinetic and potential integrals remain unchanged. Thus its positive $E_2 = 4c^2 E_{\mathrm{grad},0}$ is a geometric preparation cost; this continuum identity neither replaces the stored discrete E_2 nor constitutes a new acceptance criterion or determines the transport sign.

Compensating the initial energy through second order. A further, separately specified family keeps the same first angular variation but replaces the initial chirp by $\beta(\varepsilon) = \beta_0 + \beta_2\varepsilon^2$. On each grid, $\beta_2 = -E_2/(\partial_\beta E_0)$ is determined from the preparation energy, without using the later transport. The normalized family fixes total charge exactly and cancels the initial energy change through order ε^2; it does not assert exactly equal energies at finite deformation. Only the homogeneous radial chirp tangent is newly evolved, and its contribution is combined with the stored angular second variation. The earlier fixed-charge result remains unchanged.

On the spatially refined grid, $\beta_2 = -0.0082897517$ and the normalized net-transport derivative with respect to β is -11.0018872. Their product adds 0.0912029133 to the previous transport coefficient: $b_{QE} = 0.1785687477$, compared with $b_Q = 0.0873658344$. The base and time-refined values are 0.1785384553 and 0.1785384478; the predeclared heuristic guard for the spatially refined transport coefficient is 1.525216×10^{-4}, not a certified error interval. The initial core correction is unchanged, while the final core coefficient $c_{QE} = Q_{2,\mathrm{core},QE}(32)/Q_0$ is now $+0.0726948606$, compared with -0.0185080526 for the previous family. This final coefficient is reported separately from the net-transport decision; no additional endpoint-success gate was introduced. The compensation also reduces the initial inward radial current through its negative chirp correction. Relative to the radial reference, this energy-compensated family changes geometry and radial phase together; its transport response cannot isolate geometry alone. These coefficients concern the fixed endpoint $T = 32$ in the same finite box, not a finite-amplitude deformed solution, stationary formation or

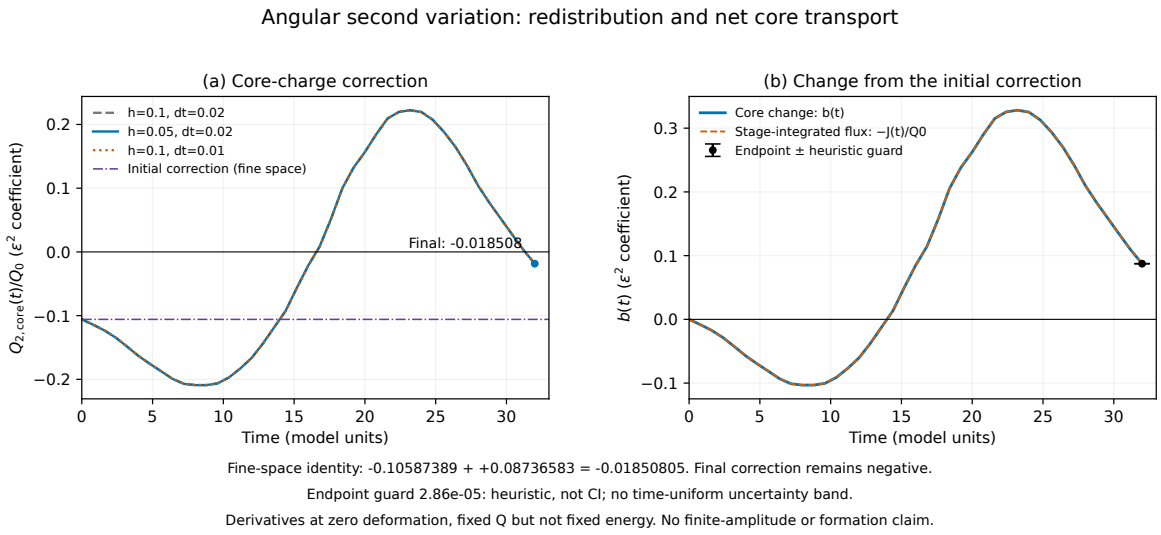


Figure S12: Second-order angular response from stored samples. Left: $Q_{2,\mathrm{core}}(t)/Q_0$ for all three discretizations, with the initial fine-space correction marked separately. Right: the initial-subtracted coefficient $b(t)$ and the negative, independently stage-integrated outward charge flux divided by Q_0, on the fine-space grid. These routes check the same charge balance, not separate physical replications. Straight lines join 41 stored samples; no finite ε is reconstructed. The final core correction remains negative despite positive net transport. The endpoint guard is heuristic, not a confidence interval, certified error or time-uniform uncertainty band.

binding. Figure S13 separates the core coefficients from their initial-subtracted transport for the two preparations.

A local two-observable time-shift test. The energy-compensated response need not describe a new stationary state: even an increased endpoint core coefficient could reflect a shifted phase of the transient concentration. A fixed stored-state check at $T = 32$ therefore compares two radial observables, $C = Q_{\mathrm{core}}/Q_0$ and $H = R^2_{\mathrm{RMS}}/6^2$, where R^2_{RMS} is the second radial moment weighted by $|\phi|^2$ and divided by its spatial integral. Write their second-order coefficients as C_2, H_2. Here $C_2(32) = c_{QE}$, not the initial-subtracted b_{QE}. A common local shift $t \mapsto t + \varepsilon^2\tau$ would require $(C_2, H_2) = \tau(\dot{C}_0, \dot{H}_0)$, hence

$$\mathcal{D} = \dot{C}_0 H_2 - \dot{H}_0 C_2 = 0.$$

The moment and its derivative were evaluated directly from the stored Cauchy data, including the norm-denominator variation; no time-series fit or additional field evolution was used. On the fine spatial grid, $H_2 = -1.0936132$ and $\mathcal{D} = 0.0021220476$, exceeding the predeclared heuristic sensitivity guard 6.820019×10^{-5} in inverse model-time units. Thus this pair is inconsistent with one common local clock shift under that diagnostic rule. The charge-only shift coefficient is $\tau_C = C_2/\dot{C}_0 = -9.1439597$, after its separate denominator check: it multiplies ε^2 and is not a measured finite time shift. This is only a two-observable endpoint comparison, conditional on the existing variational outputs; it establishes neither persistence of the core enhancement nor binding, stationary formation or a full-field distance. The sensitivity guard is not a certified error bound.

A finite ellipsoidal response at matched initial charge and energy. We additionally evolved finite, axisymmetric ellipsoidal preparations at $\varepsilon = \pm 0.02, \pm 0.04$, with initial charge and full discrete energy matched to each discretization's radial reference within the declared root tolerance. Unlike the preceding second variation, this tests finite initial data rather than energy compensation only through order ε^2. The four discretizations use the same radial box $R = 80$ and

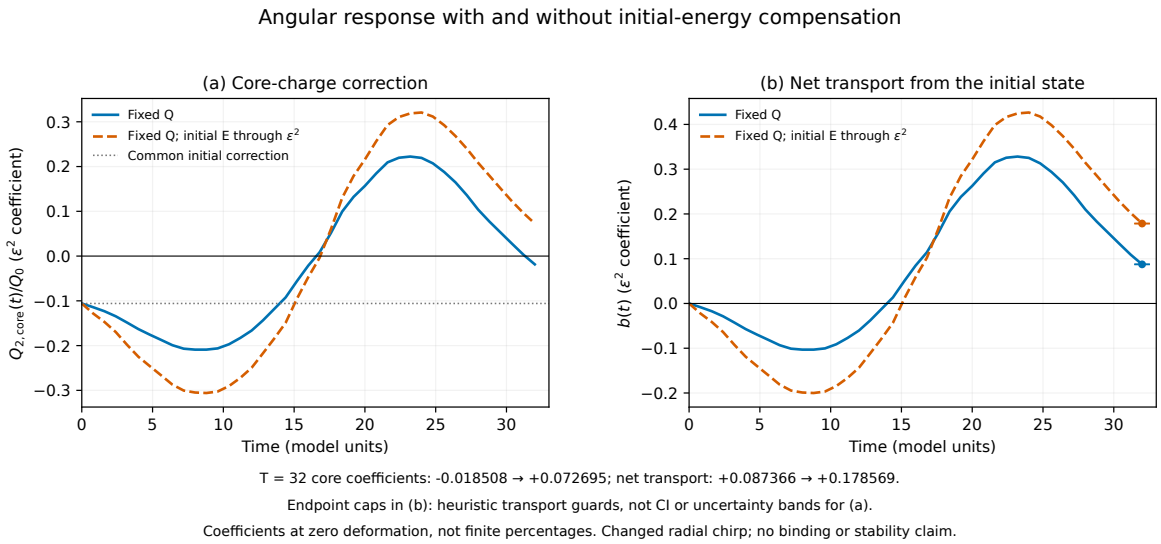


Figure S13: Stored spatially refined second-variation coefficients with and without initial-energy compensation. Left: the core-charge correction; right: its change from the common initial correction. Straight segments join 41 stored samples, without fitting or a new evolution. The original family fixes charge; the compensated family also fixes initial energy through order ε^2 while changing the radial chirp. Neither curve represents a finite deformation or a percentage increase. Endpoint caps on the transport panel are heuristic guards, not confidence intervals, core-coefficient error bars or time-uniform bounds. The sign change of the final core coefficient is not a binding or stability result.

endpoint $T = 32$:

arm	h	Δt	$\ell_{\max}$	N_θ
A	0.10	0.010	6	19
S	0.05	0.010	6	19
T	0.10	0.005	6	19
W	0.10	0.010	4	13

Only even ℓ with $m = 0$ are retained, with Gauss–Legendre angular quadrature using $N_\theta = 3\ell_{\max}+1$ nodes. Thus W changes the angular truncation rather than the radial grid. At the fixed endpoint $T = 32$ define

$$D(e) = \frac{C(+e) + C(-e) - 2C(0)}{2e^2}, \qquad C = Q_{\mathrm{core}}/Q_0.$$

This is a symmetric final core response, not initial-subtracted net transport. On the spatially refined grid, $D(0.02) = 0.0727746617$ and $D(0.04) = 0.0730141978$, against the new same-grid, same-timestep variational reference $c_{\mathrm{ref}} = 0.0726948531$. The prescribed heuristic envelope at $e = 0.02$ is 1.2055721×10^{-4}; the discrepancy 7.9808621×10^{-5} plus this allowance meets the predeclared one-percent agreement criterion. This envelope is not a certified error bound or a confidence interval. The resolved-remainder criterion is not met, so no measured Taylor order is claimed.

The separate core-fraction differences at $\varepsilon = +0.02, -0.02$ are 2.8253326×10^{-5} and 2.9966404×10^{-5}, respectively, both positive in the stored endpoint data; the symmetric envelope is not a separate certified bound for each sign. These are dimensionless charge-fraction differences, not energy percentages; a positive symmetric coefficient alone would not establish both signs. Figure S14 separates them. The preparation changes geometry and radial chirp together. This finite-time comparison establishes neither persistent enhancement, stationary formation, binding nor general three-dimensional stability. The even-ℓ, $m = 0$ evolution excludes non-axisymmetric dynamics.

The complete twenty-trajectory matrix and separate variational reference were assembled from two bounded production assignments: fourteen completed trajectories and the reference were inherited unchanged, and six remaining trajectories were completed separately. The first assignment retains its `INCOMPLETE` status; the full matrix, not a favourable subset, is used in the comparison.

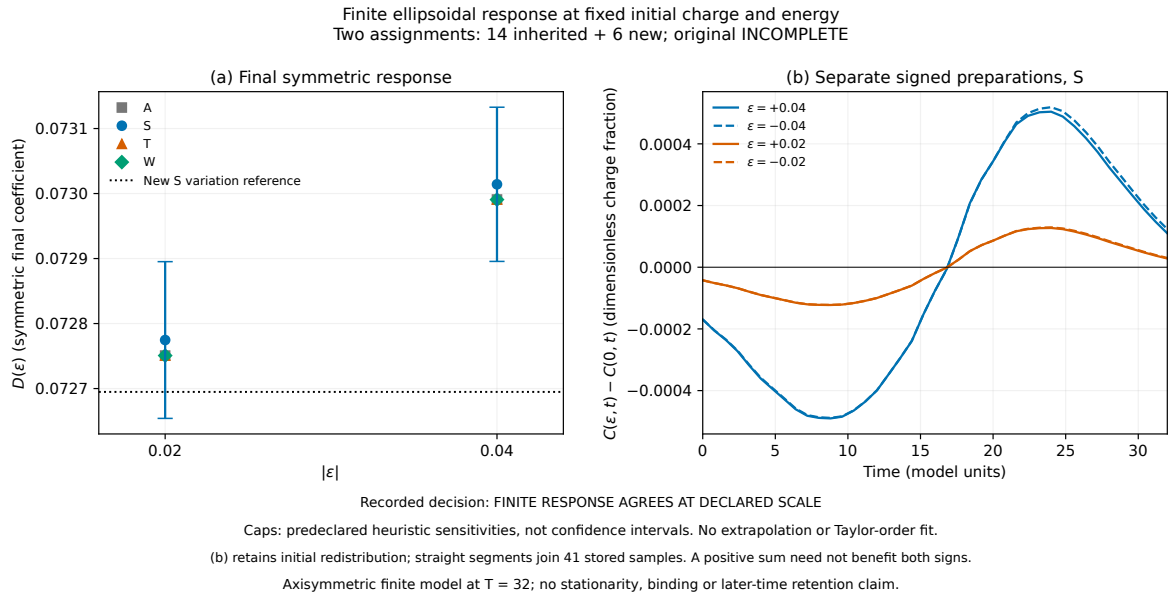


Figure S14: Finite symmetric endpoint coefficients for four discretizations (left) and separate signed core-fraction differences on the spatially refined grid (right). Caps are heuristic sensitivities, not confidence intervals. Straight segments join stored time samples; no extrapolation or Taylor-order fit is used. Initial redistribution is retained rather than subtracted.

Phase rotation and the fixed-charge kinetic gap. A similar phase colour does not test rigid harmonic motion. We evaluated necessary conditions on the six stored states with $\varepsilon = 0, \pm 0.04$ at $t = 0, 32$, using the complete $R = 80$ box. The stored reduced mode arrays use the angular convention

$$r\phi(r,\theta) = \sum_{\ell \text{ even}} U_\ell(r) B_\ell(\cos\theta), \qquad B_\ell = \sqrt{2\ell+1}\,P_\ell = \sqrt{4\pi}\,Y_{\ell 0},$$

so that $\int_{S^2} B_\ell B_{\ell'}\,\mathrm{d}\Omega = 4\pi\delta_{\ell\ell'}$. For these coefficient arrays U and $V = \partial_t U$, write $\|X\|_h^2 = 4\pi h \sum_{j=1}^{J-1}\sum_\ell |X_{j\ell}|^2$, $I = \|U\|_h^2$, $K = \|V\|_h^2$ and, for $I > 0$, $\omega_{\mathrm{fit}} = Q/(2I)$. With $A_h(U)$ the actual discrete nonlinear acceleration, define

$$r_v = \frac{\|V - \mathrm{i}\omega_{\mathrm{fit}} U\|_h^2}{K + \omega_{\mathrm{fit}}^2 I}, \qquad r_a = \frac{\|A_h(U) + \omega_{\mathrm{fit}}^2 U\|_h^2}{\|A_h(U)\|_h^2 + \omega_{\mathrm{fit}}^4 I}.$$

Rigid single-frequency motion requires both numerators to vanish. Initially r_v vanishes by construction, whereas $r_a \simeq 0.0830$; the prescribed phase velocity does not solve the stationary-profile equation. At $T = 32$ the three states give $r_v = 0.0282474$–0.0282631 and $r_a = 0.0586355$–0.0588005. These are squared-residual quotients, not amplitude percentages or kinetic-energy fractions. This completed endpoint postprocessing involved no further time integration. It provides neither a core-only classification, a relaxation law, a certified continuum residual bound nor a general 3D stability result.

The unnormalized velocity residual has a separate exact algebraic meaning. In the positive canonical kinetic norm, minimizing velocities at fixed instantaneous U and total Q gives $V_Q =$

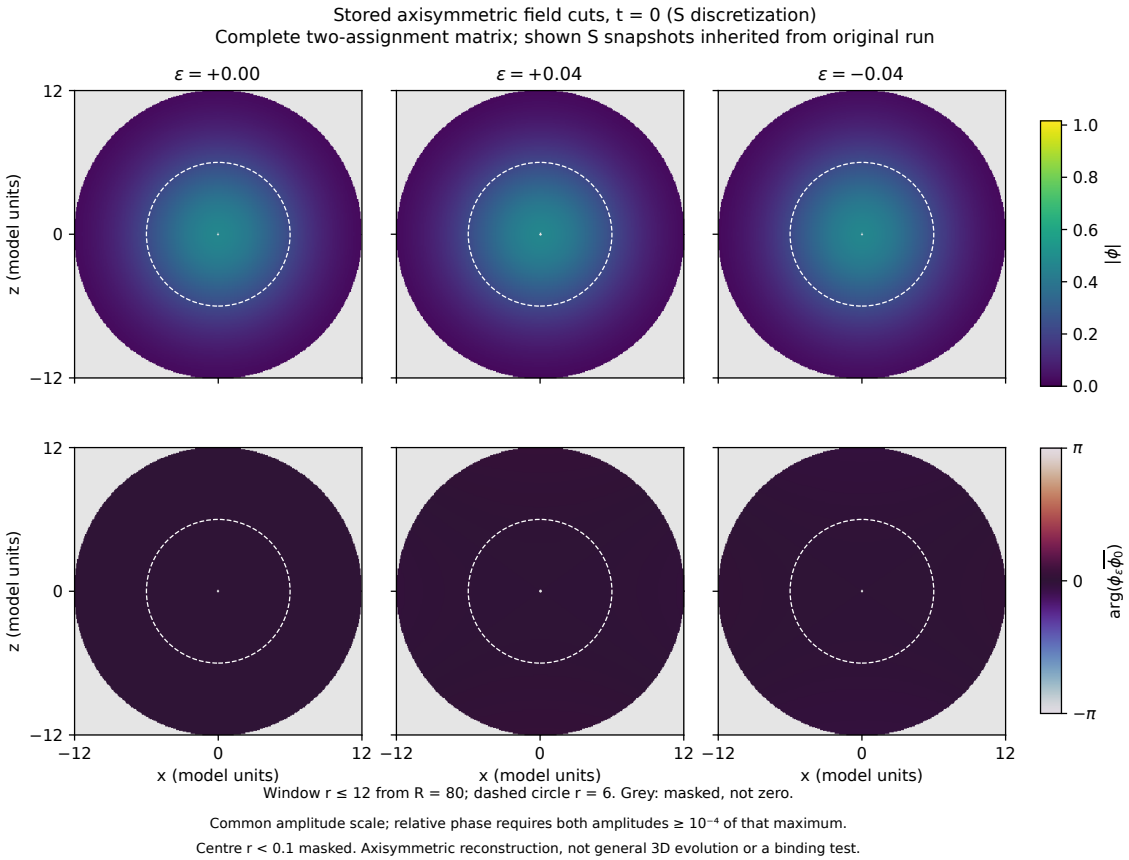


Figure S15: Stored initial amplitude (upper row) and relative phase (lower row) for the radial and $\varepsilon = \pm 0.04$ preparations. The meridional cut uses $r \leq 12$ inside the computational domain $R = 80$; the circle marks the measurement radius $r = 6$. The common amplitude scale is shared with Fig. S16. Relative phase is $\arg(\phi_\varepsilon \overline{\phi_0})$ in the common initial phase convention, masked if either amplitude is below 10^{-4} of the shared maximum. The centre $r < 0.1$ is masked; grey is not zero amplitude or phase.

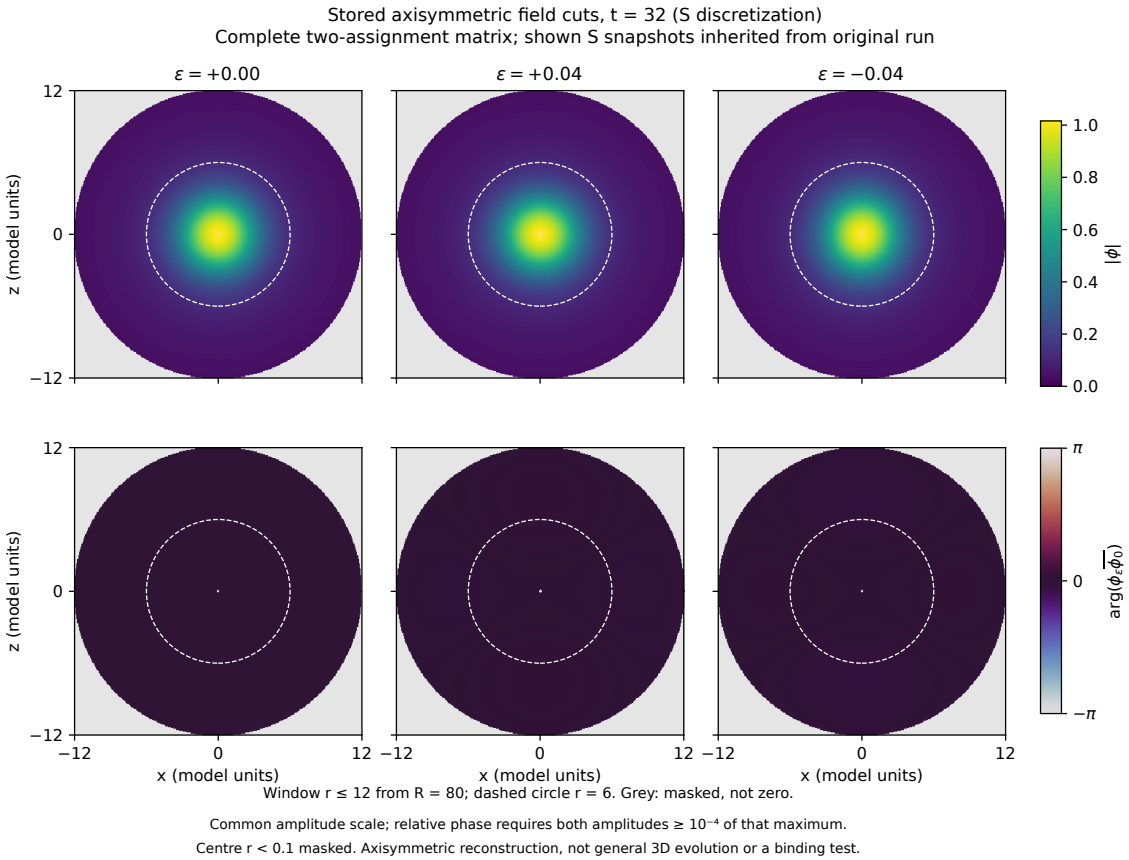


Figure S16: The same stored-field reconstruction at $T = 32$, with the common scales and masks of Fig. S15. Complex radial mode coefficients are interpolated before evaluating the angular basis and phase. These are axisymmetric amplitude/relative-phase cuts, not a general 3D simulation, an absolute-phase observable, or a binding diagnostic. Visually similar colours on the full $[-\pi, \pi]$ scale do not establish exact phase equality.

$\mathrm{i}QU/(2I)$ and

$$E_h(U,V) - E_{Q,h}[U] = J_v = \|V - \mathrm{i}\omega_{\mathrm{fit}}U\|_h^2 = K - \frac{Q^2}{4I}, \qquad E_{Q,h}[U] = W_h(U) + \frac{Q^2}{4I},$$

where W_h contains the radial and angular gradient plus potential energy. This is a comparison at fixed profile, not a physical projection or measured relaxation: it generally changes regional charge and need not preserve other conserved quantities. It does not minimize the spatial profile, show how energy is radiated, or establish binding. In particular, r_v normalizes J_v by $K + Q^2/(4I)$, not by kinetic or total energy.

Direction of the exterior field. For the same chirp, we analysed the stored radial Cauchy data $u = r\phi_b$, $v = \partial_t u$ at $t = 16, 24, 32$ on all three discretizations. A fixed window χ rises from zero to one on $12 < r < 16$ as $10s^3 - 15s^4 + 6s^5$, $s = (r - 12)/4$; $(\chi u, \chi v)$ is zero-padded onto a periodic analysis interval of length $L = 320$. With Fourier series normalized by $1/M$ for M grid points, both temporal branches are retained: $a_\sigma = (\widehat{\chi u} + \widehat{\chi v}/(\sigma \mathrm{i}\Omega))/2$, $\Omega = \sqrt{1 + k^2}$, $\sigma = \pm 1$. The convention $\exp(\mathrm{i}kr + \sigma \mathrm{i}\Omega t)$ gives group velocity $-\sigma k/\Omega$ and branch energy $8\pi L\Omega^2 |a_\sigma|^2$. The denominator E_{win} below is the *total free spectral window energy*, including the directionless $k = 0$ and unresolved Nyquist bins; it is neither the original nonlinear energy nor its escaped fraction.

The fine spatial grid gives the values below. Define $\epsilon_{\mathrm{nl}} = \|\chi(-2S_b + \frac{3}{2}S_b^2)u\|_2 / \|\chi u\|_2$ and $\epsilon_\chi = \| - 2\chi' u_r - \chi'' u\|_2 / \|\chi u\|_2$, using the radial discrete L^2 norm. These are operator-source quotients in the mass-1 units of the model (dimensionally mass squared), not relative energy errors.

t	$E_{\mathrm{out}}/E_{\mathrm{win}}$	ϵ_{nl}	ϵ_χ
16	0.904702	4.55352×10^{-5}	0.593850
24	0.888886	1.14544×10^{-4}	0.828330
32	0.939306	1.58566×10^{-3}	0.573484

All nine decompositions have $E_{\mathrm{out}} > E_{\mathrm{in}} + E_{\mathrm{Nyquist}}$ and $E_{\mathrm{out}} > E_{\mathrm{win}}/2$. The largest change in the outgoing fraction under spatial refinement is 4.59×10^{-4}, and under timestep halving 1.78×10^{-8}; these are sensitivity bounds over the sampled cases, not certified errors. The exterior nonlinear source is weak in these units, but the filter source is not: its quotient ranges from 0.57 to 0.83. Zero extension also produces the separately monitored boundary term $u_r(80^-)\delta(r - 80)$, rather than another volume norm. Thus the fixed-window data are predominantly outward-oriented; without control of the accumulated source terms, this does not establish asymptotically free radiation, stationary Q-ball formation or binding of a remaining core.

A prescribed weak-potential extension. In an exploratory, prescribed instantaneous Klein–Gordon–Poisson closure, matching candidates associated with the first two ladder labels persisted at two weak couplings each and shifted to lower ω^2. These are four tested points, not a uniform robustness region. The observation remains conditional on unresolved Poisson-residual and matching-map regularity checks; it does not establish robustness in the Einstein–Klein–Gordon system or nonlinear stability. Including the potential perturbation here means response within the same instantaneous closure, not full general relativity. The quantity $2|\Phi(0)|$ denotes twice the central potential depth and is not identified with the compactness $2GM/R$.

S6 Separate numerical-method controls

The original finite-correlation field calculation remains a numerical acceptance failure, and no accepted OU mean is inferred from it. A separate constant-source control exposed a finite-step defect in the RK4 quadratic balance: an exact finite-step RK4 identity reproduced the stored endpoint and accumulated-rate discrepancies, within the prescribed floating-point comparison tolerances, for 27 case–step combinations sampled at 31 times each. This identifies the numerical mechanism in those controls, not the complete cause of the earlier spatial-field failure. A subsequent two-stage Gauss control evaluated nine constant-system cases (three damping rates and three complex sources) at three time steps. Its recorded acceptance checks passed, including independent exact-state and directly integrated-rate references at the finest step, the prescribed error non-growth checks, and rejection of an intentionally incorrect rate. The conclusion therefore does not rest solely on Gauss's algebraic quadratic-balance property. These controls alone do not validate the original physical-pulse field calculation or decide the OU-versus-Wiener comparison; the subsequent prescribed-pulse calculation is reported separately in this supplement.

A subsequent spatial adapter used a smooth manufactured solution on the actual finite operator with $R = 20$, $h = 0.04$ and 499 interior nodes, through $T = 1$. It covered nine damping rates $\lambda = 1, 11, \ldots, 81$ and a separate zero control at time steps 0.005, 0.0025 and 0.00125. Independent full-state references and direct integral references for five original rates and three modified quadratic rates passed their finest-step accuracy checks and prescribed error non-growth checks, with a non-author result review supporting the recorded outcome. This control alone validates only the manufactured solution, not the prescribed physical pulse, an OU mean, or a continuum limit. The archive's provenance inventory identifies the separate fixed plans, results and non-author reviews for these controls, the physical-pulse comparison, its exterior sensitivity, signed lag kernel, angular second variation, compensated-energy comparison and local-clock test. These are historical numerical controls and stored-data diagnostics, rather than new physical findings or interval-certified integration errors.

S7 Archive structure and evidential scope

The companion archive is distributed with an inventory mapping original records to portable paths and SHA256 bindings distinguishing historical records from derived export copies. The three primary linear certificate runs were reproduced for preprint 0.51: their receipts agree with the originals after omitting four timing fields, and all six point and box step records are byte-identical. This does not validate a new implementation or reproduce the planar-wall comparison and nonlinear coefficient bound, which retain separate provenance. Failed sufficient checks and adaptive repetitions remain recorded with their original status.

The overview figure of the main paper (Fig. 1) is rendered from archived data and evaluations of formulas stated in the paper, with no new simulation, fit or certificate run. Its data inputs are the archive files `figures/modes/PROFILE.json` (SHA-256 prefix `9509436ad3fd5771`), `ladder/figure/figure-data.json` (`1dcf9139e23bc316`) and `ladder/extension/sprossen.json` (`4267791b7d2dfaae`), together with the midpoints of Table 1. Panel (a) also evaluates the thin-wall formula, Eq. (15), and panel (b) the function $V(f)$ of the figure caption; panel (c) uses only the certified midpoints. The phase inset, the dashed tail in panel (d) and the width curve in panel (e) are schematic. The plotting script (SHA-256 prefix `ff60bc0f6b6083d3`) reads the plotted coordinates directly from the archived fields. Before rendering it compares every plotted location with the value displayed in Table 2 to one unit of the last displayed digit, derives each evidence class from that table's evidence column, and compares the certified midpoints with Table 1. These are consistency checks of the rendering, not additional evidence for the physics. Its receipt records the full checksums of script, inputs and outputs. Script, receipt and input copies are published in the folder `fig-overview` of the preprint 0.52 sources; they are not part of the 0.51 archive. In the published copies, three lines of the script's

header text and the host field of the receipt are replaced by neutral wording; a map there lists the checksums of the run copies and of the published copies.

References

[1] S. Coleman, "Q-balls," *Nuclear Physics B* **262**, 263–283 (1985), doi:10.1016/0550-3213(85)90286-X; erratum, *Nuclear Physics B* **269**, 744 (1986), doi:10.1016/0550-3213(86)90520-1.

[2] A. Kovtun, E. Nugaev, and A. Shkerin, "Vibrational modes of Q-balls," *Physical Review D* **98**, 096016 (2018), doi:10.1103/PhysRevD.98.096016, arXiv:1805.03518v2.

[3] D. Ciurla, P. Dorey, T. Romańczukiewicz, and Y. Shnir, "Perturbations of Q-balls: from spectral structure to radiation pressure," *Journal of High Energy Physics* **07** (2024), 196, doi:10.1007/JHEP07(2024)196, arXiv:2405.06591v2.

[4] H. Friedrich and D. Wintgen, "Interfering resonances and bound states in the continuum," *Physical Review A* **32**, 3231–3242 (1985), doi:10.1103/PhysRevA.32.3231.

[5] X. Yu and Y. Y. Lu, "Existence of Friedrich–Wintgen bound states in the continuum: system of Schrödinger equations," *SIAM Journal on Applied Mathematics* **86**, 1654–1671 (2026), doi:10.1137/25M1733926, arXiv:2504.19573v1 (2025).

[6] S. Kasuya and M. Kawasaki, "Q-ball formation through the Affleck–Dine mechanism," *Physical Review D* **61**, 041301 (2000), doi:10.1103/PhysRevD.61.041301, arXiv:hep-ph/9909509v3.

[7] A. Kusenko and M. Shaposhnikov, "Supersymmetric Q-balls as dark matter," *Physics Letters B* **418**, 46–54 (1998), doi:10.1016/S0370-2693(97)01375-0, arXiv:hep-ph/9709492v3.

[8] A. Azatov, Q. T. Ho, and M. M. Khalil, "Q-ball perturbations with more details: linear analysis vs lattice," *Physical Review D* **111**, 096010 (2025), doi:10.1103/PhysRevD.111.096010, arXiv:2412.13885v1 (2024).

[9] G.-D. Zhang, S.-Y. Zhou, and M.-F. Zhu, "*Q*-ball superradiance: Analytical approach," *Physical Review D* **113**, 056020 (2026), doi:10.1103/rwj6-nsl8, arXiv:2510.27064v1 (2025).

[10] H.-Y. Zhang, M. A. Amin, E. J. Copeland, P. M. Saffin, and K. D. Lozanov, "Classical Decay Rates of Oscillons," *Journal of Cosmology and Astroparticle Physics* **07** (2020), 055, arXiv:2004.01202v2 (2020), doi:10.1088/1475-7516/2020/07/055.

[11] J. Evslin, H. Liu, T. Romańczukiewicz, Y. Shnir, A. Wereszczyński, and P. Ziobro, "Linearized Q-Ball Perturbations," arXiv:2604.07713v1 (2026).

[12] B. A. Malomed, T. Wagenknecht, A. R. Champneys, and M. J. Pearce, "Accumulation of embedded solitons in systems with quadratic nonlinearity," *Chaos* **15**, 037116 (2005), doi:10.1063/1.1938433, arXiv:nlin/0505012.

[13] A. Soffer and M. I. Weinstein, "Resonances, Radiation Damping and Instability in Hamiltonian Nonlinear Wave Equations," *Inventiones mathematicae* **136**, 9–74 (1999), doi:10.1007/s002220050303.

[14] R. A. Battye and P. M. Sutcliffe, "Q-ball Dynamics," *Nuclear Physics B* **590**, 329–363 (2000), doi:10.1016/S0550-3213(00)00506-X, arXiv:hep-th/0003252v1.

[15] F. Johansson, "Arb: Efficient Arbitrary-Precision Midpoint-Radius Interval Arithmetic," *IEEE Transactions on Computers* **66**(8), 1281–1292 (2017), doi:10.1109/TC.2017.2690633.

[16] The Python–FLINT developers, *Python–FLINT 0.9.0 documentation*, https://python-flint.readthedocs.io/en/latest/ (accessed 1 October 2026).

[17] The FLINT developers, *FLINT: Fast Library for Number Theory*, official repository, https://github.com/flintlib/flint (accessed 1 October 2026).

[18] R. Krawczyk, "Newton-Algorithmen zur Bestimmung von Nullstellen mit Fehlerschranken," *Computing* **4**, 187–201 (1969), doi:10.1007/BF02234767.

[19] K. Koshelev, S. Lepeshov, M. Liu, A. Bogdanov and Y. Kivshar, "Asymmetric metasurfaces with high-Q resonances governed by bound states in the continuum," *Physical Review Letters* **121**, 193903 (2018), doi:10.1103/PhysRevLett.121.193903, arXiv:1809.00330.

[20] S. Flach, A. E. Miroshnichenko, V. Fleurov, and M. V. Fistul, "Fano resonances with discrete breathers," Phys. Rev. Lett. **90**, 084101 (2003), doi:10.1103/PhysRevLett.90.084101, arXiv:cond-mat/0211313.

[21] S. Flach, V. Fleurov, A. V. Gorbach, and A. E. Miroshnichenko, "Resonant light scattering by optical solitons," Phys. Rev. Lett. **95**, 023901 (2005), doi:10.1103/PhysRevLett.95.023901, arXiv:nlin/0409023.

[22] S. Watabe, Y. Kato, and Y. Ohashi, "Excitation Transport through a Domain Wall in a Bose–Einstein Condensate," Phys. Rev. A **86**, 023622 (2012), doi:10.1103/PhysRevA.86.023622, arXiv:1208.0381.

[23] J. Heeck, A. Rajaraman, R. Riley, and C. B. Verhaaren, "Understanding Q-Balls Beyond the Thin-Wall Limit," Phys. Rev. D **103**, 045008 (2021), doi:10.1103/PhysRevD.103.045008, arXiv:2009.08462.

[24] T. Inagaki and Y. Murakami, "Open-channel radiation zeros and nonlinear damping of a critical-bubble internal mode," arXiv:2609.15056 (2026).

[25] J. L. Marzuola and G. Simpson, "Spectral analysis for matrix Hamiltonian operators," *Nonlinearity* **24**, 389–429 (2011), doi:10.1088/0951-7715/24/2/003, arXiv:1003.2474.

[26] R. Asad and G. Simpson, "Embedded eigenvalues and the nonlinear Schrödinger equation," *Journal of Mathematical Physics* **52**, 033511 (2011), doi:10.1063/1.3567152, arXiv:1101.2485.

[27] C. W. Hsu, B. Zhen, A. D. Stone, J. D. Joannopoulos, and M. Soljačić, "Bound states in the continuum," *Nature Reviews Materials* **1**, 16048 (2016), doi:10.1038/natrevmats.2016.48.

[28] H. Segur and M. D. Kruskal, "Nonexistence of small-amplitude breather solutions in ϕ^4 theory," *Physical Review Letters* **58**, 747–750 (1987), doi:10.1103/PhysRevLett.58.747; erratum, **58**, 1158 (1987), doi:10.1103/PhysRevLett.58.1158.2.

[29] I. M. Sigal, "Non-linear wave and Schrödinger equations. I. Instability of periodic and quasiperiodic solutions," *Communications in Mathematical Physics* **153**, 297–320 (1993), doi:10.1007/BF02096645.

[30] R. Friedberg, T. D. Lee, and A. Sirlin, "Class of scalar-field soliton solutions in three space dimensions," *Physical Review D* **13**, 2739–2761 (1976), doi:10.1103/PhysRevD.13.2739.

[31] N. G. Vakhitov and A. A. Kolokolov, "Stationary solutions of the wave equation in a medium with nonlinearity saturation," *Radiophysics and Quantum Electronics* **16**, 783–789 (1973), doi:10.1007/BF01031343.

[32] Q. Chen, L. Andersson, and L. Li, "Stability analysis for Q-balls with spectral method," arXiv:2509.18656v1 (2025).

[33] S. Cuccagna and M. Maeda, "A survey on asymptotic stability of ground states of nonlinear Schrödinger equations II," arXiv:2009.00573v1 (2020).

[34] D. Li and K. Yang, "The linearized cubic NLS has no embedded eigenvalue," *Inventiones mathematicae* **245**, 879–975 (2026), doi:10.1007/s00222-026-01419-3.

[35] J. Lührmann, J. M. Palacios, F. Pusateri, W. Schlag, and S. Shahshahani, "Asymptotic stability of the degree-one vortex in the abelian Yang–Mills–Higgs model: Spectral Theory and Numerics," arXiv:2608.10608v1 (2026).

[36] B. Zhen, C. W. Hsu, L. Lu, A. D. Stone, and M. Soljačić, "Topological nature of bound states in the radiation continuum," *Physical Review Letters* **113**, 257401 (2014), doi:10.1103/PhysRevLett.113.257401, arXiv:1408.0237v1.

[37] N. S. Manton and H. Merabet, "ϕ^4 Kinks—Gradient Flow and Dynamics," *Nonlinearity* **10** (1997), doi:10.1088/0951-7715/10/1/002, arXiv:hep-th/9605038v2 (1996).